The discrete-time arbitrage-free Nelson–Siegel model: a closed-form solution and applications to mixed funds representation*

Ramin Eghbalzadeh[a], Frédéric Godin[†,a,b], and Patrice Gaillardetz[a,b]

[a]Concordia University, Department of Mathematics and Statistics, Montréal, Canada

[b]Quantact Laboratory, Centre de Recherches Mathématiques, Montréal, Canada

March 19, 2023

Abstract

A closed-form solution for zero-coupon bonds is obtained for a version of the discrete-time arbitrage-free Nelson-Siegel model. An estimation procedure relying on a Kalman filter is provided. The model is shown to produce adequate fit when applied to historical Canadian spot rate data and to improve distributional predictive performance over benchmarks. An adaptation of the mixed fund return model from Augustyniak et al. (2021) is also provided to include the discrete-time arbitrage-free Nelson-Siegel model as one of its building blocks.

JEL classification: E43, G12.

Keywords: Interest rate models, Term structure, Arbitrage-free Nelson-Siegel, Gaussian factor models, Kalman filter, Mixed funds.

*We thank the Natural Sciences and Engineering Research Council of Canada (Gaillardetz: RGPIN-2020-06821, Godin: RGPIN-2017-06837) and the Canadian Institute of Actuaries for their financial support.

†Corresponding author.

Email addresses: r.nirvan@gmail.com (Ramin Eghbalzadeh), patrice.gaillardetz@concordia.ca (Patrice Gaillardetz), frederic.godin@concordia.ca (Frédéric Godin).

1 Introduction

Dynamics of interest rates have implications in different areas such as asset and derivatives pricing, portfolio management, and risk measurement. In the context of actuarial science, interest rate models are central to the evaluation of long-term financial contracts such as standard insurance and annuities, as well as equity-linked products. Superior models provide enhanced predictive power and risk evaluation ability, which are crucial to insurers and pensioners. Interest rate models often study the dynamics of the term structure specifying simultaneously spot rates for different tenors. The evolution of the term structure can be represented by various types of models, which can be of statistical, economic, or financial nature. Statistical approaches, such as these of Diebold and Li (2006), Guidolin and Timmermann (2006) or Engle et al. (2017), mainly rely on time series models. In economic methods, see for instance Ang and Piazzesi (2003), Hördahl et al. (2006) or Ang et al. (2011), the evolution of the term structure is linked to macro-economic factors, such as inflation rates, the growth rate of the economy and the central bank monetary policy. Financial approaches are based on the absence of arbitrage paradigm ensuring that pricing systems are consistent. Brigo and Mercurio (2007) is a comprehensive reference describing multiple models from that category.

Vasicek (1977) and Cox et al. (1978) are early contributions in the stream of literature of financial models for the term structure. They propose to model the evolution of the short rate as specific Itô diffusions leading to a closed-form solution for zero-coupon prices and the corresponding spot rates. Multi-factor generalizations of such models leading to more flexible term structure shapes and to the ability to properly model correlations between future spot rates have then been proposed. See for instance Brennan and Schwartz (1979), Chen and Scott (1992), Rogers (1995) and Jamshidian (1996) for example of multi-factor models of the term structure. Factor models can be motivated for instance by the work of Litterman and Scheinkman (1991) showing that at least 95% of variation in the yield curve can be explained by three statistical factors. An important category of financial factor models for interest rates that is described in Duffie and Kan (1996) is affine term structure (ATS) models, in which any spot rate is a linear combination of underlying factors. Such models are typically obtained by assuming diffusive dynamics for the factors and deriving corresponding bond prices through common no-arbitrage arguments. By suitably choosing the drift of the factors, it is possible to obtain a model-implied term structure that exactly match the observed one, see for instance Hull and White (1990). The latter model is a particular case of the Heath et al. (1992) framework which, instead of directly modeling the short rate, uses the current term structure as initial input and applies perturbations on it to represent the evolution of the yield curve.

An issue with some multi-factor models of the term structure is that the factors are sometimes hard to interpret. The seminal work of Nelson and Siegel (1987) and the Svensson (1994) extension specifically aim to depict the term structure as a linear combination of interpretable factors representing the level, slope and humps in the yield curve. Whereas the latter two models were initially developed in a static setting, Diebold and Li (2006) extend the Nelson and Siegel (1987) model into a dynamic framework. However, a drawback associated with the dynamic version of the Nelson and Siegel (1987) model is that it is shown to be incompatible with no-arbitrage theory in Filipović (1999). The work of Christensen et al. (2011) successfully attempts to reconcile both the no-arbitrage strand of literature and the Nelson and Siegel (1987) model by developing the so-called arbitrage-free Nelson-Siegel (AFNS) model. Such model is an affine term structure model where the specification of the diffusion followed by term structure factors is carefully chosen such that factor loadings exactly coincide with these of the Nelson-Siegel model. The discrepancy between spot rates of the Nelson-Siegel model and of the AFNS model only stems from a corrective term whose inclusion is necessary to preclude theoretical arbitrage opportunities. There are debates in the literature about whether enforcing absence of arbitrage in interest rate models is necessary for predictive purposes: Duffee (2002) argues that ATS models often have a poor performance when used for forecasting future Treasury yields. Nevertheless, when considering asset pricing applications, enforcing absence of arbitrage is often required to obtain a meaningful model.

Many of the aforementioned financial models of the term structure consider continuous-time settings. However, most of these models have analog discrete-time counterparts, many of which are described in Wüthrich and Merz (2013). Discrete-time models often possess the advantage of being more simple to manipulate, which is favorable in applications. A discrete-time multi-factor Gaussian term structure model is illustrated in Augustyniak et al. (2021), in which it serves as a building block for an econometric model for mixed equity and fixed income fund returns. That paper uses the mixed fund model to price variable annuities. The discrete-time analog of the multi-factor Cox-Ingersoll-Ross model is developed in the work of Le et al. (2010), which uses auto-regressive Gamma processes introduced by Gouriéroux and Jasiak (2006). A discrete-time version of the AFNS model has been developed in Hong et al. (2016). As mentioned above, the AFNS model possesses both the favorable attributes of being consistent with no-arbitrage considerations and of embedding interpretable risk factors; having a discrete-time version of such model available is therefore highly relevant for practitioners who desire avoiding technicalities associated with continuous-time models and who wish to retain positive features of the AFNS model.

This paper further expands on the work Hong et al. (2016) and revisits the discrete-time arbitrage-free Nelson-Siegel (DTAFNS) model, providing the following three main contributions: (i) obtaining a closed-form expression for risk-free spot rates under such model, (ii) presenting evidence of superior distributional out-of-sample predictive ability of the DTAFNS model over benchmarks when applied on historical Canadian term structure data, and (iii) developing a modified version of the mixed fund return model of Augustyniak et al. (2021) by substituting their three-factor Gaussian model with the DTAFNS model. The latter is justified by the higher performance of the DTAFNS model outlined in this paper.

The remainder of this paper is organized as follows. Section 2 specifies the proposed interest rate model and provides closed-form expressions for risk-free zero-coupon bond prices and spot rates. In Section 3, the model calibration is illustrated and numerical experiments assessing predictive performance are provided. Section 4 adapts the mixed fund model of Augustyniak et al. (2021) to include the DTAFNS model as a building block for the term structure model. Section 5 concludes.

2 Interest rate term structure model

This section presents the mathematical construction of the discrete-time version of the arbitrage-free Nelson-Siegel term structure model, referred to subsequently as the DTAFNS model. First, the specification of the traditional Nelson-Siegel model and its dynamic continuous-time arbitrage-free extension are recalled. Then the dynamics of the short rate in the DTAFNS model is provided and closed-form formulas for spot rates are derived. As explained in more details subsequently, the DTAFNS model presented in this section has a slighlty different specification from that of Hong et al. (2016).

2.1 The Nelson-Siegel term structure representation

Nelson and Siegel (1987) propose a parametric representation of the yield curve of the form

$$y(t,T) = X^{(1)} + X^{(2)}\left(\frac{1-e^{-\lambda\tau}}{\lambda\tau}\right) + X^{(3)}\left(\frac{1-e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau}\right), \tag{2.1}$$

where $y(t,T)$ is the time-t spot rate with tenor $\tau = T - t$, and $X^{(1)}, X^{(2)}, X^{(3)}$ and $\lambda > 0$ are model parameters. The model leads to a convenient interpretation of the parameters, with $X^{(1)}$, $X^{(2)}$ and $X^{(3)}$ respectively characterizing the level, slope and curvature of the yield curve, while λ is a dilation parameter. Although this model has first been considered in a static setting, Diebold and Li (2006) embed it in a dynamic setting where time-varying factors $X_t^{(1)}, X_t^{(2)}, X_t^{(3)}$ are considered instead of $X^{(1)}, X^{(2)}, X^{(3)}$. Several auto-regressive models are used for the dynamics of such parameters, thereby making it possible to forecast the term structure.

A pitfall of the dynamic Nelson-Siegel model of Diebold and Li (2006) is its incompatibility with no-arbitrage theory as discussed Filipović (1999). Christensen et al. (2011) therefore adapt the Nelson-Siegel framework and develop the AFNS model, an arbitrage-free term structure model whose spot rate formulas match exactly these of the Nelson-Siegel model up to a correction term. It is obtained by specifying the short rate risk-neutral dynamics as the following affine Gaussian

Itô diffusion:

$$
\begin{aligned}
r_t &\equiv X_t^{(1)} + X_t^{(2)}, && (2.2)\\
\begin{bmatrix} dX_t^{(1)} \\ dX_t^{(2)} \\ dX_t^{(3)} \end{bmatrix} &\equiv \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & \lambda & -\lambda \\ 0 & 0 & \lambda \end{bmatrix}}_{\kappa^{\mathbb{Q}}} \underbrace{\begin{bmatrix} \theta_1^{\mathbb{Q}} - X_t^{(1)} \\ \theta_2^{\mathbb{Q}} - X_t^{(2)} \\ \theta_3^{\mathbb{Q}} - X_t^{(3)} \end{bmatrix}}_{\theta^{\mathbb{Q}} - \mathbf{X}_t} dt + \Sigma \begin{pmatrix} dW_{t,1}^{\mathbb{Q}} \\ dW_{t,2}^{\mathbb{Q}} \\ dW_{t,3}^{\mathbb{Q}} \end{pmatrix} && (2.3)
\end{aligned}
$$

where $\{W_{t,1}^{\mathbb{Q}}\}_{t\geq 0}$, $\{W_{t,2}^{\mathbb{Q}}\}_{t\geq 0}$ and $\{W_{t,3}^{\mathbb{Q}}\}_{t\geq 0}$ are independent Brownian motions under the risk-neutral probability measure $\mathbb{Q}$, Σ is a 3×3 positive semi-definite matrix and $\theta_1^{\mathbb{Q}}, \theta_2^{\mathbb{Q}}, \theta_3^{\mathbb{Q}}, \lambda$ are real numbers. As shown in Christensen et al. (2011), this leads to the following formulas for spot rates:

$$
\begin{aligned}
y(t,T) &= -\frac{1}{\tau} \sum_{j=1}^{3} \tilde{B}^{(j)}(t,T) X_t^{(j)} - \frac{\tilde{C}(t,T)}{\tau}, && (2.4)\\
\tilde{C}(t,T) &= \sum_{j=2}^{3} \left(\kappa^{\mathbb{Q}} \theta^{\mathbb{Q}}\right)_j \int_t^T \tilde{B}^{(j)}(s,T) ds + \frac{1}{2} \sum_{j=1}^{3} \int_t^T \left(\Sigma^\top \tilde{B}(s,T) \tilde{B}(s,T)^\top \Sigma\right)_{j,j} ds && (2.5)
\end{aligned}
$$

with

$$
\begin{aligned}
\tilde{B}(s,T) &\equiv \left[\tilde{B}^{(1)}(s,T),\ \tilde{B}^{(2)}(s,T),\ \tilde{B}^{(3)}(s,T)\right]^\top \\
&\equiv \left[-\tau,\ -\left(\frac{1-e^{-\lambda\tau}}{\lambda}\right),\ \left(\tau e^{-\lambda\tau} - \frac{1-e^{-\lambda\tau}}{\lambda}\right)\right]^\top && (2.6)
\end{aligned}
$$

and $()_j$ or $()_{j,j}$ denoting respectively the jth element of a vector or the element at row j and column j of a matrix, respectively.

Striking similarities can be observed when comparing (2.1) with (2.4) and (2.6). Indeed, identifying $X_t^{(1)}, X_t^{(2)}, X_t^{(3)}$ with $X^{(1)}, X^{(2)}, X^{(3)}$, discrepancies between the spot rates from both models is only caused by the corrective term $-\frac{\tilde{C}(t,T)}{\tau}$ appearing in (2.4). This leads Christensen et al. (2011) to describe the arbitrage-free Nelson-Siegel model (2.2)-(2.3) as the "*closest match to the Nelson-Siegel yield function*" within some class of exponential affine term structure models.

2.2 Discrete-time arbitrage-free Nelson-Siegel model

The purpose of this section is to define a discrete-time model analogous to model (2.2)-(2.3). In what follows, an arbitrage-free market with discrete time steps $t = 0, 1, ..., T$, each separated by a time elapse Δ, is considered. Market dynamics are defined on probability space $(\Omega, \mathcal{F}_T, \mathbb{P})$ where $\mathbb{P}$ is the real-world probability measure and $\mathcal{F} := \{\mathcal{F}_t\}_{t=0}^{T}$ is a filtration characterizing the information flow in the market.

In the discrete-time context, the time-t short risk-free rate r_t is the $\mathcal{F}_t$-measurable rate effective for the period $[t, t+1)$. The short-rate dynamics considered are

$$r_t = X_t^{(1)} + X_t^{(2)}, \tag{2.7}$$

$$X_{t+1} = X_t + \kappa^{\mathbb{P}}(\theta^{\mathbb{P}} - X_t) + \Sigma Z_{t+1}^{\mathbb{P}}, \tag{2.8}$$

where the stochastic factor vectors are $X_t = \left[X_t^{(1)}, X_t^{(2)}, X_t^{(3)}\right]^\top$, $t = 0, \ldots, T$, $\kappa^{\mathbb{P}}$ and Σ are constant parameter matrices of dimension 3×3, and $\theta^{\mathbb{P}}$ is a constant column vector of dimension 3. The process $Z^{\mathbb{P}} = \{Z_t^{\mathbb{P}}\}_{t=1}^{T}$ with $Z_t^{\mathbb{P}} = \left[Z_{t,1}^{\mathbb{P}}, Z_{t,2}^{\mathbb{P}}, Z_{t,3}^{\mathbb{P}}\right]^\top$ is a multivariate standard Gaussian white noise under $\mathbb{P}$ with contemporaneous correlation parameters $\rho_{i,j}^{(Z)} \equiv \text{corr}(Z_{t,i}^{\mathbb{P}}, Z_{t,j}^{\mathbb{P}})$, $t = 1, \ldots, T$ and $i, j = 1, 2, 3$. The initial value of the factor process is fixed at $X_0 = x_0 \in \mathbb{R}^3$. Σ is assumed to be a diagonal matrix with strictly positive elements. Equations (2.7)-(2.8) are meant to mimic (2.2)-(2.3), except that the model is specified under the physical measure $\mathbb{P}$ instead of under the risk-neutral measure $\mathbb{Q}$. A more general class of Gaussian discrete-time affine structure models is studied in Wüthrich and Merz (2013), but without specializing to the DTAFNS model which is the object of this paper.

To obtain the specification of spot rates under the DTAFNS model, risk-neutral dynamics need to be specified. Indeed, spot rates are given by

$$y(t, T) = -\frac{\log P(t, T)}{(T - t)\Delta}$$

where $P(t, T)$ is the time-t of a risk-free zero coupon of unit face value with maturity on time point T. In discrete-time models of the short rate, the relationship between $P(t,T)$ and the risk-free rate is

$$P(t,T) = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\Delta\sum_{j=t}^{T-1} r_j\right)\bigg|\mathcal{F}_t\right].$$

To obtain risk-neutral dynamics, following the lines of Augustyniak et al. (2021) for instance, a discrete-time version of the Girsanov theorem can be invoked to apply a translation to the drift of short rate factors X_t without altering their volatilities, while keeping a multivariate Gaussian distribution for innovations. It follows from this theorem that for a given constant market price of risk matrix $\gamma \in \mathbb{R}^{3\times 3}$, there exists a probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ such that the process $Z^{\mathbb{Q}} = \{Z_t^{\mathbb{Q}}\}_{t=1}^{T}$ defined through $Z_{t+1}^{\mathbb{Q}} = Z_{t+1}^{\mathbb{P}} - \gamma X_t$ is also an $\mathcal{F}$-adapted standard Gaussian white noise under $\mathbb{Q}$, still with contemporaneous correlation matrix $\rho^{(Z)} = \left[\rho_{i,j}^{(Z)}\right]_{i,j=1}^{3}$. Substituting $Z_{t+1}^{\mathbb{P}} = Z_{t+1}^{\mathbb{Q}} + \gamma X_t$ into (2.8) leads to

$$X_{t+1} = X_t + \kappa^{\mathbb{P}}\theta^{\mathbb{P}} - (\kappa^{\mathbb{P}} - \Sigma\gamma)X_t + \Sigma Z_{t+1}^{\mathbb{Q}}.$$

Assuming there is a solution $(\kappa^{\mathbb{Q}}, \theta^{\mathbb{Q}})$ with $\kappa^{\mathbb{Q}} \in \mathbb{R}^{3\times 3}$ and $\theta^{\mathbb{Q}} \in \mathbb{R}^{3\times 1}$ to the system of equations

$$\begin{cases} \kappa^{\mathbb{Q}}\theta^{\mathbb{Q}} = \kappa^{\mathbb{P}}\theta^{\mathbb{P}}, \\ \kappa^{\mathbb{Q}} = \kappa^{\mathbb{P}} - \Sigma\gamma, \end{cases} \tag{2.9}$$

then

$$X_{t+1} = X_t + \kappa^{\mathbb{Q}}(\theta^{\mathbb{Q}} - X_t) + \Sigma Z_{t+1}^{\mathbb{Q}} \tag{2.10}$$

and thus the factor process $\{X_t\}_{t=0}^{T}$ preserves its auto-regressive structure under the risk-neutral measure.

The DTAFNS model relies on specific choices of parameters mimicking the AFNS model. Zero-coupon bond prices under such specific choices are hereby derived. The proofs for the next results are provided in Appendix A.

Lemma 2.1. *For some integer $\tau > 0$ and a real number $r \neq 1$,*

$$\zeta_0(r,\tau) \equiv \sum_{u=1}^{\tau-1} r^u = \frac{r - r^\tau}{1-r}, \tag{2.11}$$

$$\zeta_1(r,\tau) \equiv \sum_{u=1}^{\tau-1} u r^u = \frac{r - \tau r^\tau + (\tau-1) r^{\tau+1}}{(1-r)^2}, \tag{2.12}$$

$$\zeta_2(r,\tau) \equiv \sum_{u=1}^{\tau-1} u^2 r^u = \frac{-(\tau-1)^2 r^{\tau+2} + (2\tau^2 - 2\tau - 1) r^{\tau+1} - \tau^2 r^\tau + r^2 + r}{(1-r)^3}. \tag{2.13}$$

Proposition 2.1. *Consider $\lambda \in (0,1)$ and assume*

$$\kappa^{\mathbb{Q}} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \lambda & -\lambda \\ 0 & 0 & \lambda \end{pmatrix}. \tag{2.14}$$

In other words, suppose that the state variables X_t have the following dynamics risk-neutral dynamics:

$$\begin{pmatrix} X_{t+1}^{(1)} - X_t^{(1)} \\ X_{t+1}^{(2)} - X_t^{(2)} \\ X_{t+1}^{(3)} - X_t^{(3)} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \lambda & -\lambda \\ 0 & 0 & \lambda \end{pmatrix} \begin{bmatrix} \theta_1^{\mathbb{Q}} - X_t^{(1)} \\ \theta_2^{\mathbb{Q}} - X_t^{(2)} \\ \theta_3^{\mathbb{Q}} - X_t^{(3)} \end{bmatrix} + \begin{pmatrix} \Sigma_{1,1} & 0 & 0 \\ 0 & \Sigma_{2,2} & 0 \\ 0 & 0 & \Sigma_{3,3} \end{pmatrix} \begin{pmatrix} Z_{t+1,1}^{\mathbb{Q}} \\ Z_{t+1,2}^{\mathbb{Q}} \\ Z_{t+1,3}^{\mathbb{Q}} \end{pmatrix}.$$

Then, the time-t arbitrage-free price of a zero-coupon maturing at T is, for $t = 0, \ldots, T-1$,

$$P(t,T) = A_\tau \exp\left[-\Delta B_\tau^\top X_t\right], \tag{2.15}$$

where $\tau = T - t$, $B_\tau = \left[B_\tau^{(1)},\ B_\tau^{(2)},\ B_\tau^{(3)}\right]^\top$ and

$$B_\tau^{(1)} = \tau, \quad B_\tau^{(2)} = \frac{1-(1-\lambda)^\tau}{\lambda}, \quad B_\tau^{(3)} = \frac{1-(1-\lambda)^{\tau-1}}{\lambda} - (\tau-1)(1-\lambda)^{\tau-1},$$

$$\log A_\tau = -\Delta\theta_2^{\mathbb{Q}}\left(B_\tau^{(1)} - B_\tau^{(2)}\right) + \Delta\theta_3^{\mathbb{Q}} B_\tau^{(3)} + \frac{1}{2}\Delta^2 \upsilon_\tau$$

with

$$
\begin{aligned}
\upsilon_\tau &= \sum_{i=1}^{3}\sum_{j=1}^{3}\upsilon_\tau^{(i,j)}\\
\upsilon_\tau^{(1,1)} &= \Sigma_{1,1}^2\frac{\tau(\tau-1)(2\tau-1)}{6},\\
\upsilon_\tau^{(2,2)} &= \frac{\Sigma_{2,2}^2}{\lambda^2}\left(\tau-2\left[\frac{1-(1-\lambda)^\tau}{\lambda}\right]+\frac{1-(1-\lambda)^{2\tau}}{1-(1-\lambda)^2}\right),\\
\upsilon_\tau^{(3,3)} &= \frac{\Sigma_{3,3}^2}{\lambda^2}\Bigg[\tau-2+\zeta_0\left((1-\lambda)^2,\tau-1\right)+\lambda^2\zeta_2\left((1-\lambda)^2,\tau-1\right)\\
&\quad-2\zeta_0\left((1-\lambda),\tau-1\right)-2\lambda\zeta_1\left((1-\lambda),\tau-1\right)+2\lambda\zeta_1\left((1-\lambda)^2,\tau-1\right)\Bigg],\\
\upsilon_\tau^{(1,2)} &= \upsilon_\tau^{(2,1)}=\rho_{1,2}\Sigma_{1,1}\Sigma_{2,2}\frac{1}{\lambda}\left(\frac{\tau(\tau-1)}{2}-\zeta_1((1-\lambda),\tau)\right),\\
\upsilon_\tau^{(1,3)} &= \upsilon_\tau^{(3,1)}=\rho_{1,3}\Sigma_{1,1}\Sigma_{3,3}\frac{1}{\lambda}\Bigg[\frac{\tau(\tau-1)}{2}-1-\zeta_0\left((1-\lambda),\tau-1\right)-(\lambda+1)\zeta_1\left((1-\lambda),\tau-1\right)\\
&\quad-\lambda\zeta_2\left((1-\lambda),\tau-1\right)\Bigg],\\
\upsilon_\tau^{(2,3)} &= \upsilon_\tau^{(3,2)}=\rho_{2,3}\Sigma_{2,2}\Sigma_{3,3}\Bigg(\frac{\tau-2-(2-\lambda)\zeta_0\left((1-\lambda),\tau-1\right)+(1-\lambda)\zeta_0\left((1-\lambda)^2,\tau-1\right)}{\lambda^2}\\
&\quad+\frac{-\zeta_1\left((1-\lambda),\tau-1\right)+(1-\lambda)\zeta_1\left((1-\lambda)^2,\tau-1\right)}{\lambda}\Bigg).
\end{aligned}
$$

This entails that spot rates have the form

$$
y(t,T)=X_t^{(1)}+\frac{(1-(1-\lambda)^\tau)}{\tau\lambda}X_t^{(2)}+\left[\frac{(1-(1-\lambda)^{\tau-1})}{\tau\lambda}-\frac{\tau-1}{\tau}(1-\lambda)^{\tau-1}\right]X_t^{(3)}-\frac{1}{\Delta\tau}\log A_\tau. \tag{2.16}
$$

Remark 2.1. *In Proposition 2.1, parameter $\theta_1^{\mathbb{Q}}$ does not appear in the zero-coupon price formula (2.15) and thus can be set to $\theta_1^{\mathbb{Q}}=0$.*

To obtain risk-neutral dynamics of the form proposed in Proposition 2.1, the following physical dynamics is considered:

$$
\kappa^{\mathbb{P}}=\begin{pmatrix}\kappa_{1,1}^{\mathbb{P}} & 0 & 0\\ 0 & \kappa_{2,2}^{\mathbb{P}} & -\lambda\\ 0 & 0 & \kappa_{3,3}^{\mathbb{P}}\end{pmatrix},\quad \theta^{\mathbb{P}}=\begin{pmatrix}0\\ \theta_2^{\mathbb{P}}\\ \theta_3^{\mathbb{P}}\end{pmatrix}
$$

which allows meeting the second condition in (2.9) with

$$\gamma = \begin{pmatrix} \gamma_1 & 0 & 0 \\ 0 & \gamma_2 & 0 \\ 0 & 0 & \gamma_3 \end{pmatrix}, \quad \gamma_i = (\kappa^{\mathbb{P}}_{i,i} - \mathbb{1}_{\{i>1\}}\lambda)/\Sigma_{i,i},\ i = 1, 2, 3. \tag{2.17}$$

Furthermore, substituting (2.17) in the first equation of (2.9) leads to

$$\theta^{\mathbb{Q}}_2 = \lambda^{-1}(\kappa^{\mathbb{P}}_{2,2}\theta^{\mathbb{P}}_2 + \kappa^{\mathbb{P}}_{3,3}\theta^{\mathbb{P}}_3 - \lambda\theta^{\mathbb{P}}_3), \quad \theta^{\mathbb{Q}}_3 = \lambda^{-1}\kappa^{\mathbb{P}}_{3,3}\theta^{\mathbb{P}}_3.$$

Conversely, physical parameters can be retrieved from risk-neutral parameters through

$$\kappa^{\mathbb{P}}_{i,i} = \mathbb{1}_{\{i>1\}}\lambda + \Sigma_{i,i}\gamma_i,\ i = 1, 2, 3, \quad \theta^{\mathbb{P}}_3 = \lambda\theta^{\mathbb{Q}}_3/\kappa^{\mathbb{P}}_{3,3}, \quad \theta^{\mathbb{P}}_2 = \frac{\lambda}{\kappa^{\mathbb{P}}_{2,2}}\left[\theta^{\mathbb{Q}}_2 - \frac{\theta^{\mathbb{Q}}_3}{\kappa^{\mathbb{P}}_{3,3}}(\kappa^{\mathbb{P}}_{3,3} - \lambda)\right]. \tag{2.18}$$

Remark 2.2. *Since $\kappa^{\mathbb{Q}}_{1,1} = 0$, an alternative specification enforcing $\kappa^{\mathbb{P}}_{1,1} = 0$ has been tested, but it was ultimately not retained due to lower performance. See Appendix D of the Online Appendix for more details.*

Remark 2.3. *The DTANFS model of this paper is specified slightly differently than in Hong et al. (2016). In the latter work, the matrix $\kappa^{\mathbb{Q}}$ is chosen such that factor loadings B_τ exactly match that of the Nelson-Siegel model (2.1). Conversely, this paper emphasizes an auto-regressive representation of factors, i.e. Equation (2.10), which is the discrete-time counterpart of the diffusive dynamics of the AFNS model (2.3). Nevertheless, both specifications of the model are quite similar conceptually. In Hong et al. (2016), only a recursive representation of A_τ is provided, which makes the derivation of the closed-form solution for A_τ a useful contribution of the present work.*

3 Model estimation

This section presents a method to estimate the DTAFNS model using historical spot curves data. The model is estimated on Canadian interest rate data. The fit performance is analyzed and benchmarked against the discrete-time Gaussian three-factor model of Augustyniak et al. (2021) and a version of the Dynamic Nelson-Siegel model of Diebold and Li (2006).

3.1 Estimation framework

Several methods have been considered in the literature to estimate factor models of interest rates, such as conventional maximum likelihood estimation (Chen and Scott, 1993), maximum likelihood with Kalman filters (Duan and Simonato, 1999; Lemke, 2006; Park, 2014; Augustyniak et al., 2021), the method of simulated moments (Dai and Singleton, 2000) or Bayesian approaches (Hong et al., 2016). In this paper, the Kalman filter approach is applied. Observed spot rates (observable data) are considered to be a noisy version of the true spot rates (the signal). Setting up the Kalman filter requires to first derive the recursive Gaussian linear relationship between the observable quantities and factors driving the signal, which is hereby provided.

Assume that on each time t a set of M annualized continuously compounded spot rates with fixed times-to-maturity $n_1, ..., n_M$ are observed, which are denoted by the column vector

$$\hat{\mathsf{y}}(t) \equiv (\hat{y}(t, t+n_1), ..., \hat{y}(t, t+n_M))^\top.$$

Define also the corresponding model-implied spot rate vector

$$\mathsf{y}(t) \equiv (y(t, t+n_1), ..., y(t, t+n_M))^\top$$

where (2.16) leads to

$$y(t, t+n) = \frac{-1}{n\Delta} \log A_n + \frac{1}{n} B_n^\top X_t. \tag{3.1}$$

The relationship between observed and model implied spot rate is assumed to be

$$\hat{\mathsf{y}}(t) = \mathsf{y}(t) + \eta_{\mathsf{t}}$$

where $\{\eta_t\}_{t=1}^T$ is a multivariate Gaussian white noise with diagonal variance-covariance matrix H.

Furthermore, for simplicity, all values on the diagonal of matrix H are assumed to be identical and equal to some parameter h. This leads to

$$\hat{\mathsf{y}}(t) = \mathsf{a} + \mathsf{B}X_t + \eta_t, \tag{3.2}$$

where

$$\begin{aligned}
\mathsf{a} &\equiv \left(\frac{-1}{n_1\Delta}\log A_{n_1}, \ldots, \frac{-1}{n_M\Delta}\log A_{n_M}\right)^{\top}, && (3.3)\\
\beta_n &\equiv \left(\frac{B_n^{(1)}}{n}, \frac{B_n^{(2)}}{n}, \frac{B_n^{(3)}}{n}\right)^{\top}, \\
\mathsf{B} &\equiv \left(\beta_{n_1}, \ldots, \beta_{n_M}\right)^{\top}. && (3.4)
\end{aligned}$$

Furthermore, the transition equation describing the dynamics of the latent factors X_t is obtained from (2.8):

$$X_{t+1} = \mathbf{b} + \mathbf{D}X_t + \xi_{t+1}, \tag{3.5}$$

where

$$\mathbf{b} \equiv \kappa^{\mathbb{P}}\theta^{\mathbb{P}}, \quad \mathbf{D} \equiv I - \kappa^{\mathbb{P}}, \quad \xi_{t+1} \equiv \Sigma Z_{t+1}^{\mathbb{P}}. \tag{3.6}$$

The sequence $\{\xi_t\}_{t=1}^{T}$ is therefore a multivariate Gaussian white noise with covariance matrix

$$\mathbf{Q} = \Sigma\rho\Sigma. \tag{3.7}$$

The representation (3.2)-(3.5) of state space variables allows estimating the DTAFNS model using a Kalman filter. The general Kalman filter algorithm is presented in Appendix C of the Online Appendix. The Kalman filter allows deriving the log-likelihood of observed spot rate curves for a candidate set of parameters. Algorithm 1 indicates steps to calculate the log-likelihood function in the context of the DTAFNS model. In this algorithm, p denotes joint density functions. In the present work, the implementation of the Kalman filter is performed with the `R` package `FKF`.

Moreover, to find maximum likelihood estimates of the parameters, consistently with Augustyniak et al. (2021), an optimization of the likelihood is conducted with the `R` package `Rsolnp` (Ghalanos and Theussl, 2015) applying the nonlinear augmented Lagrange multiplier optimization method of Ye (1987). The optimization is performed under the following constraints:

$$\Sigma_{i,i} \geq 0, \quad h \geq 0, \quad -1 \leq \rho_{i,j} \leq 1,\ i,j = 1,2,3, \quad \lambda \in (0,1).$$

Additional constraints could be applied to enforce the positive semi-definiteness of matrices Σ and ρ, but optimization results obtained with our dataset satisfy such constraints without explicitly specifying them to the optimizer.

Algorithm 1 Kalman filter algorithm for the calculation of likelihood function and smoothed state densities

---

Input: Initial values $\bar{x}_{1|0} \in \mathbb{R}^3$, $P_{1|0} \in \mathbb{R}^{3\times 3}$, parameters h, λ, $\theta^{\mathbb{Q}}$, ρ, γ, Σ and spot rate dataset $\hat{\mathsf{y}}(t)$, $t = 0, \ldots, T$.
Outputs: Log-likelihood $L(\hat{y}) = \log\left(p\left(\hat{\mathsf{y}}(T), \hat{\mathsf{y}}(T-1), ..., \hat{\mathsf{y}}(1)\right)\right)$ and smoothed density moments $\bar{x}_{t|T}, P_{t|T}$, $t = 1, \ldots, T$.

Calculating the log-likelihood:
Calculate $\mathsf{a}, \mathbf{b}, \mathbf{D}, \mathbf{Q}$ *and* B *with* (2.18), (3.3), (3.4), (3.6), (3.7) *and Proposition* 2.1.
for $t \in \{1, ..., T\}$ **do**
  Calculate $p(\hat{\mathsf{y}}(t)|\hat{\mathsf{y}}(1), ..., \hat{\mathsf{y}}(t-1))$:
  $\bar{y}(t) \leftarrow \mathsf{a} + \mathsf{B}\bar{x}_{t|t-1}$
  $\bar{\Sigma}_t \leftarrow \mathsf{B}P_{t|t-1}\mathsf{B}^\top + H$
  $p(\hat{\mathsf{y}}(t)|\hat{\mathsf{y}}(1), ..., \hat{\mathsf{y}}(t-1)) \sim N(\bar{y}(t), \bar{\Sigma}_t)$

  Calculate $\bar{x}_{t|t}$ *and* $P_{t|t}$:
  $\bar{x}_{t|t} \leftarrow \bar{x}_{t|t-1} + P_{t|t-1}\mathsf{B}^\top\bar{\Sigma}_t^{-1}(\hat{\mathsf{y}}(t) - \bar{\mathsf{y}}(t))$
  $P_{t|t} \leftarrow P_{t|t-1} - P_{t|t-1}\mathsf{B}^\top\bar{\Sigma}_t^{-1}\mathsf{B}P_{t|t-1}$
  $\bar{x}_{t+1|t} \leftarrow \mathbf{b} + \mathbf{D}\bar{x}_{t|t}$
  $P_{t+1|t} \leftarrow \mathbf{D}P_{t|t}\mathbf{D}^\top + \mathbf{Q}$
$L(\hat{y}) \leftarrow \sum_{t=1}^{T} \log p\left(\hat{\mathsf{y}}(t)|\hat{\mathsf{y}}(1), ..., \hat{\mathsf{y}}(t-1)\right)$

Calculating the smoothed state density:
for $t \in \{T-1, \ldots, 1\}$ **do**
  $\mathbf{J}_t \leftarrow P_{t|t}\mathbf{D}^\top P_{t+1|t}^{-1}$,
  $\bar{x}_{t|T} \leftarrow \bar{x}_{t|t} + \mathbf{J}_t(\bar{x}_{t+1|T} - \bar{x}_{t+1|t})$,
  $P_{t|T} \leftarrow P_{t|t} + \mathbf{J}_t(P_{t+1|T} - P_{t+1|t})\mathbf{J}_t^\top$.

The smoothed state density is the density of time-t factors conditional on all observations from the sample, namely $p(X_t|\hat{y}(1),\ldots,\hat{y}(T))$. Such distribution is multivariate Gaussian with mean $\bar{x}_{t|T}$ and covariance matrix $P_{t|T}$. Such mean and variance are obtained with the Kalman smoother algorithm (Shumway and Stoffer, 2017) which is also shown in Algorithm 1.

3.2 Model estimation and analysis

This section presents the DTAFNS model estimation using historical Canadian spot rate data.

3.2.1 The dataset

To estimate the model, end-of-month Canadian spot rate curves from January 1986 to January 2022 (434 months) are considered. Such yield curves are provided publicly on the website of the Bank of Canada[1] and are constructed through an exponential smoothing methodology described in Bolder et al. (2004). 33 spot rate tenors are considered, which include short-end maturities of 3, 6, and 9 months and all integer times-to-maturity from 1 to 30 years.[2]

3.2.2 Estimated parameters and performance

Table 1 gives estimates of DTAFNS model parameters obtained with Algorithm 1 which applies the Kalman filter. Factor 1 innovations are negatively correlated with these of the two other factors, which are positively correlated between themselves. The negative correlation between the level factor $X^{(1)}$ and the slope and curvature factors $X^{(2)}$ and $X^{(3)}$ tends to reduce the frequency of very large or low values of the short rate r_t. Furthermore, it implies that when the long-term rates increase through higher values of $X^{(1)}$, the slope given by $-X^{(2)}$ also tends to go up. Thus, movements in the long-end of the curve are not necessarily matched by movements of a similar magnitude and direction in the short-end. The hump in the curve reflected by $X^{(3)}$ also tends to decrease during periods of increasing of long-term spot rates. The speed of reversion $\kappa^{\mathbb{P}}_{i,i}$ is much lower for the first factor ($i = 1$) than for the other two, which is not surprising due to the overall rates level having had a clear lasting downward trend in the historical sample, representing close-to-non-stationary behavior. Finally, the estimation implies a long-term average of short rates of $\theta^{\mathbb{P}}_2 = 0.0301$ under physical dynamics.

[1]Source: https://www.bankofcanada.ca/rates/interest-rates/bond-yield-curves/

[2]For all years up to 1990 inclusively, times-to-maturity between 26 years and 30 years are missing in the dataset and are therefore not considered.

Table 1: Maximum likelihood estimates of the DTAFNS model parameters

i	$\kappa_{i,i}^{\mathbb{P}}$	γ_i	$\Sigma_{i,i}$	$\theta_i^{\mathbb{P}}$	$\theta_i^{\mathbb{Q}}$	$\bar{x}_{i,1\|0}$	λ	h
1	0.0075	2.7923	0.0027	0	0	0.0491		
2	0.0288	1.2016	0.0045	0.0301	0.0633	0.0391	0.0233	3.76×10^{-6}
3	0.0354	1.7167	0.0070	0.0505	0.0766	0.0291		

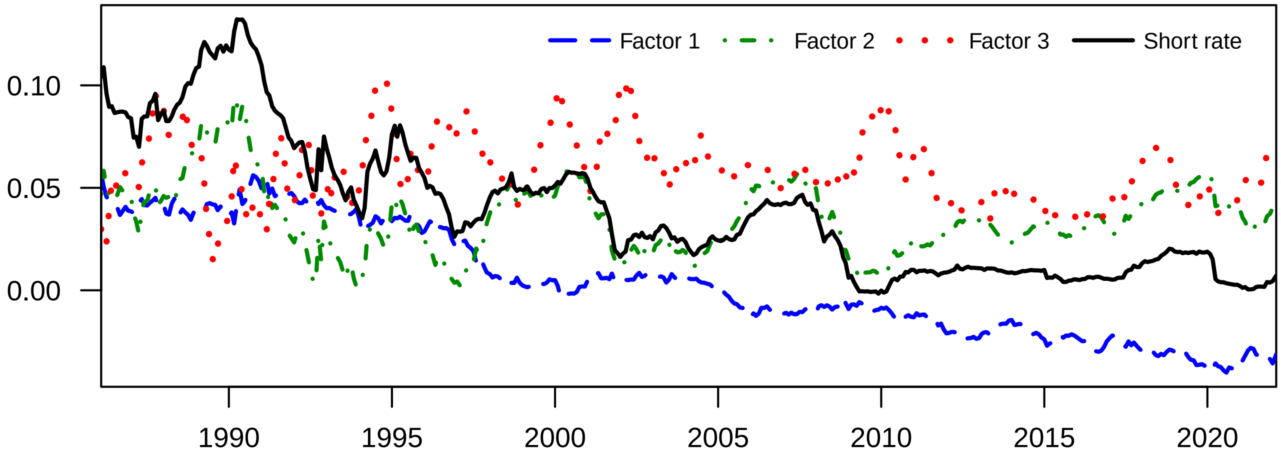
$$\rho = \begin{bmatrix} 1 & -0.6303 & -0.4097 \\ -0.6303 & 1 & 0.2993 \\ -0.4097 & 0.2993 & 1 \end{bmatrix}, \; P_{1|0} = \begin{bmatrix} 4.45\times10^{-6} & 0 & 0 \\ 0 & 4.45\times10^{-6} & 0 \\ 0 & 0 & 4.45\times10^{-6} \end{bmatrix}$$

Notes: Maximum likelihood parameter estimates for the DTAFNS model presented in Section 2.2 obtained with the Kalman filter (see Algorithm 1). The data sample includes Canadian end-of-month yield curves from January 1986 to January 2022. $\bar{x}_{i,1|0}$ refers to element i of the vector $\bar{x}_{1|0}$.

Figure 1 shows the smoothed value (the smoothed distribution expected value) of latent factors provided by the Kalman smoother procedure described in Algorithm 1. The black curve is the short rate implied by the model which is obtained by summing the smoothed values of the first two factors, consistently with (2.7). The decreasing trend of the short rate throughout the data sample is recovered by the downward trend of Factor 1. Conversely, Factor 2 and Factor 3, driving the slope and humps in the curve respectively, exhibit more stationary fluctuations.

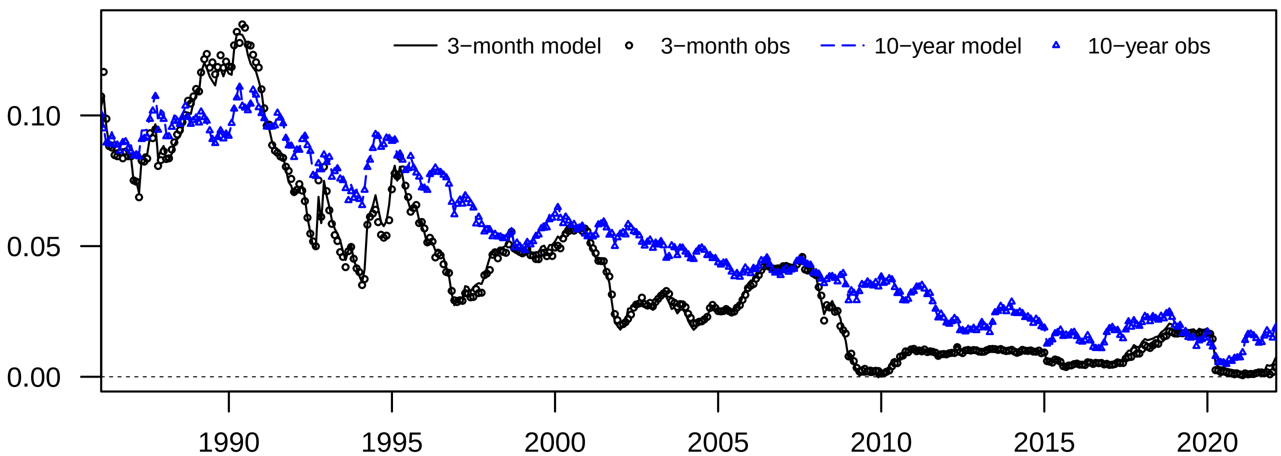
To assess the ability of the DTAFNS model to match observed short- and long-term spot rates of the term structure, time series of observed spot rates of three-month and 10-year tenors are compared to the model-implied spot rates for the same tenors. Such time series are provided in Figure 2. The model can be seen to adequately match short- and long-term rates throughout the entire sample, as differences between the model-implied and observed spot rates are very small.

Figure 1: Model-implied factors and short rate time series



Notes: Time series of DTAFNS model-implied factors which correspond to the smoothed state inferences $E^{\mathbb{P}}\left[x_t^{(i)}|\hat{y}(1), \hat{y}(2), ..., \hat{y}(T)\right]$, $i = 1, 2, 3$ provided by Algorithm 1, and implied short rates obtained by summing smoothed values of the first two factors. The model is estimated on the end-of-month Canadian spot rate curves extending from January 1986 to January 2022.

Figure 2: Model-implied and observed spot rate time series for 3-month and 10-year tenors



Notes: Time series of observed 3-month (short-term) and 10-year (long-term) maturity spot rates (dotted curves), with corresponding spot rates implied by the fitted DTAFNS model. The dataset considered is the end-of-month Canadian spot rate curves extending from January 1986 to January 2022.

3.3 Performance assessment and benchmarking

This section analyzes the adequacy of the fit and the predictive performance of the DTAFNS model through benchmarking.

3.3.1 *Benchmarks*

The discrete-time Gaussian three-factor model of Augustyniak et al. (2021), subsequently referred to as the DG3 model, and a version of the dynamic Nelson-Siegel (DNS) model introduced by Diebold and Li (2006) are benchmarks considered in this study for comparison of performance results. See Appendix B for a formal specification of these two models. The main differences between DG3 and the DTAFNS model is that (i) the latent factors in the former only regress on themselves instead of allowing cross-factor auto-regressions, (ii) the DTAFNS has a short rate equal to the sum of two of the factors instead of the three as in the benchmark, and (iii) the DTAFNS has a fixed structure for the matrix $\kappa^{\mathbb{Q}}$ driving the auto-regression intensity, unlike the Augustyniak et al. (2021) model allowing optimizing such quantities. Regarding the second benchmark, unlike in the DTAFNS model, the DNS model does not include a deterministic adjustment term in spot rates to rule out arbitrage opportunities.

3.3.2 *Goodness-of-fit of the model and forecasting ability*

To assess the model's ability to adequately fit the observed data, we compare the log-likelihood of the DTAFNS, the DG3 and the DNS models. Two types of log-likelihoods are computed: an in-sample version obtained by fitting the models to the entire dataset, and an out-of-sample version which is obtained by considering an expanding window sequential approach. For the latter, in the first iteration, data from January 1986 up to the end of 2016 is considered as a first training set, and data from 2017 is considered as a test set. In any following iteration, the training sets are expanded by one year and the test year moves to the year following that of the previous iteration. The last iteration has a test set including data for both 2021 and 2022 since the dataset only includes data for January in 2022. The aggregated out-of-sample log-likelihood is obtained by summing these associated with all test sets. Table 2 provides the results. Whereas the in-sample log-likelihood of the DG3 is higher than that of the DTAFNS and the DNS models, the out-of-sample value for the DTAFNS model is higher than that of the DNS and DG3 benchmarks for the aggregate of all test years. This implies better distributional predictive ability for the DTAFNS model.

Table 2: Log-likelihood of the DTAFNS model and its benchmarks

Model	Out-of-sample						In-sample
	2017	2018	2019	2020	2021-22	Aggregated	
DTAFNS	2,034	2,001	2,035	2,015	2,183	10,268	65,702
DG3	2,012	1,978	2,024	2,027	2,200	10,241	66,069
DNS	1,985	1,982	1,993	2,001	2,163	10,124	64,610

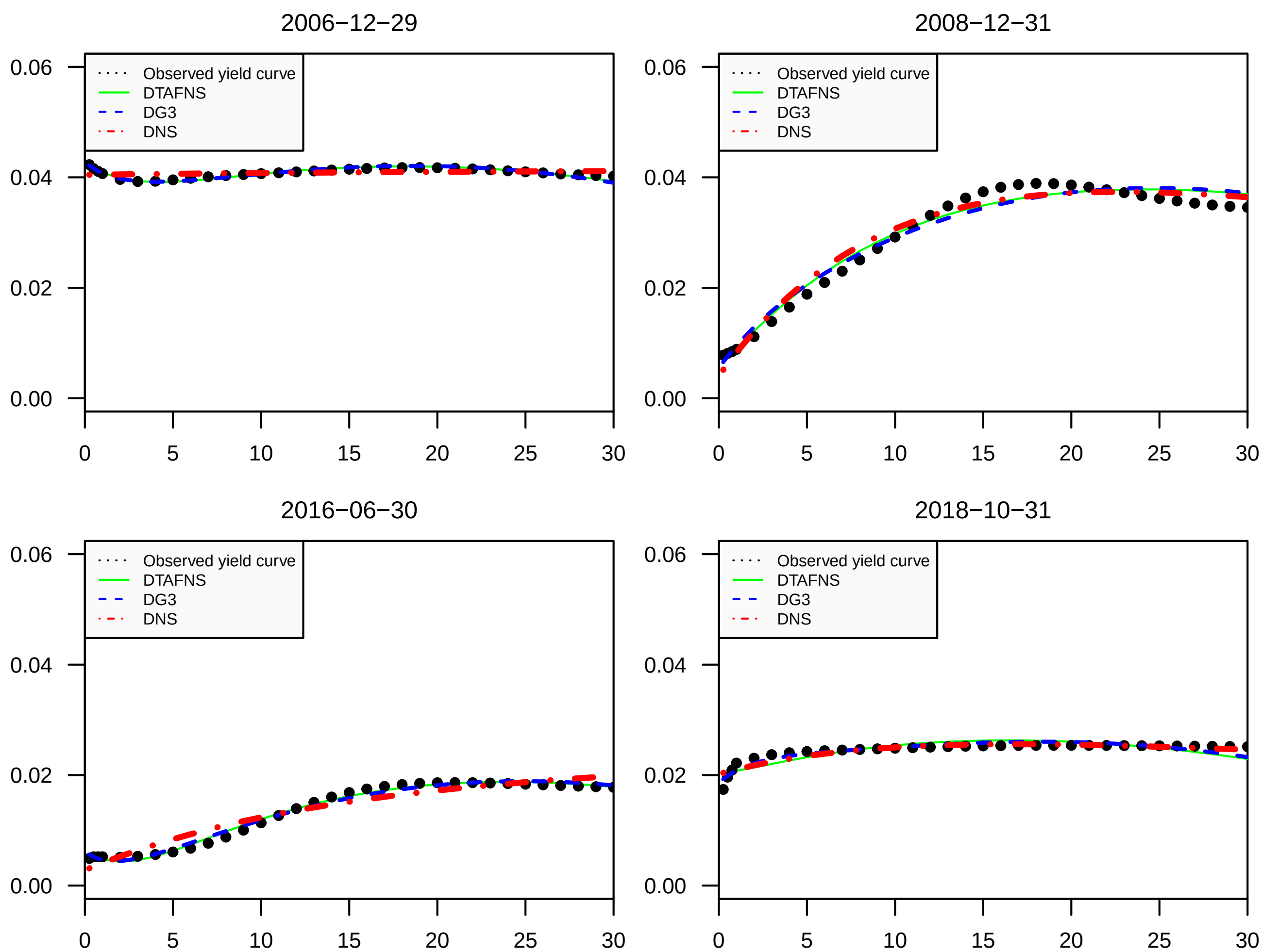
Notes: Comparison of the log-likelihood of the DTAFNS model of Section 2.2 and of the DG3 and DNS benchmarks described in Appendix B. The data sample is the Canadian end-of-month yield curve. The in-sample dataset starts in January 1986 and ends in January 2022. The out-of-sample estimation procedure uses an expanding window approach described in Section 3.3.2. The aggregated out-of-sample log-likelihood is ultimately obtained by summing the log-likelihood for all test years.

To visualize the ability of the DTAFNS model and benchmarks to replicate observed yield curves, Figure 3 presents the realized spot curves and model-implied counterparts for four selected dates. These four days are the same that are considered in Augustyniak et al. (2021) to display fitting performance, namely December 29, 2006, December 31, 2008, June 30, 2016 and October 31, 2018. Such dates exhibit several spot curve shapes: flat, upward sloping or humped. The figure shows that the DTAFNS and both benchmarks are able to reproduce observed yield curves reasonably well.

3.3.3 Frequency of negative values

A drawback associated with the use of the Gaussian distribution within interest rate models is the possibility of producing negative short rates. Although negative rates have been observed in some European markets, such phenomenon almost did not occur in North America. To investigate the propensity of the DTAFNS model and its benchmarks to generate negative interest rates, 200,000 five-year monthly paths are simulated under the physical measure with the estimated parameters drawn from Table 1, Table 6 and Table 7, respectively, and starting values for factors being their smoothed values on January 2022.

Figure 3: Model-implied and observed yield curves



Notes: Realized and model-implied spot rate curves on the four following dates: December 29, 2006, December 31, 2008, June 30, 2016 and October 31, 2018. Dotted black line: observed spot rates. Full green line: DTAFNS model implied curve. Dashed blue line: DG3 benchmark implied curve. Red dotted-dashed line: DNS benchmark implied curve. The observed data are end-of-month Canadian spot rates provided by the Bank of Canada.

Table 3 reports (i) the proportion of simulated observations below the thresholds 0, −0.01, −0.02, or −0, 03, and (ii) the proportion of paths with at least one observation below such thresholds. The DTAFNS model significantly reduces such proportions when compared with both the DG3 and DNS models, which is desirable.

Table 3: Probability of observing negative short rates with the DTAFNS model or its benchmarks

Model	Proportion of months				Proportion of paths with at least one month			
	< 0	< -0.01	< -0.02	< -0.03	< 0	< -0.01	< -0.02	< -0.03
DTAFNS	0.177	0.057	0.015	0.003	0.644	0.299	0.106	0.029
DG3	0.425	0.176	0.058	0.015	0.942	0.629	0.303	0.109
DNS	0.240	0.082	0.022	0.005	0.730	0.379	0.142	0.040

Notes: Within 200,000 five-year monthly simulated paths of the DTAFNS model and the DG3 and DNS benchmarks described in Appendix B, the proportion of (i) simulated months with short rates being smaller than respectively 0, −0.01, −0.02 and −0.03, and (ii) simulated paths with at least one month below such thresholds are reported. Model parameters are drawn from Table 1, Table 6, and Table 7, respectively, with starting values of the factors being their smoothed values on January 2022.

4 An application to mixed fund modeling

One of the main building blocks of the mixed fund (a fund containing both fixed income and equity) model of Augustyniak et al. (2021), which is used for variable annuity pricing, is the underlying factor-based DG3 interest rate model. Since the DTAFNS model is shown herein to improve predictive performance over the DG3 benchmark, it is relevant to adapt the Augustyniak et al. (2021) mixed fund model by replacing the DG3 model with the DTAFNS model.

4.1 Rolling bond fund returns

The dynamics of the mixed fund model of Augustyniak et al. (2021) involves a regression on movements of interest rate factors, among other quantities. The inclusion of such regressors is justified by an analysis of returns of a rolling bond fund, which are shown to depend on such drivers. Therefore, to adapt the Augustyniak et al. (2021), the rolling bond fund dynamics is derived under the DTAFNS dynamics.

Consider a fund containing a single zero-coupon of time-to-maturity τ that is rolled-over on each period. Denoting its time-$t+1$ log-return by $R_{t+1}^{(\tau)}$, (2.15) leads to

$$\begin{aligned} R_{t+1}^{(\tau)} &= \log\left(\frac{P(t+1,t+\tau)}{P(t,t+\tau)}\right) \\ &= \log A_{\tau-1} - \log A_{\tau} - \Delta B_{\tau-1}^{\top} X_{t+1} + \Delta B_{\tau}^{\top} X_t \\ &= \log A_{\tau-1} - \log A_{\tau} - \Delta \sum_{i=1}^{3} B_{\tau-1}^{(i)} \left(X_{t+1}^{(i)} - \left(\frac{B_{\tau}^{(i)}}{B_{\tau-1}^{(i)}}\right) X_t^{(i)} \right). \end{aligned} \tag{4.1}$$

Moreover, recalling that $r_t = X_t^{(1)} + X_t^{(2)}$ and using (2.14), (2.18), Lemma A.2 and (4.1), the excess return can be expressed as

$$\begin{aligned} R_{t+1}^{(\tau)} - \Delta r_t &= \log A_{\tau-1} - \log A_{\tau} - \Delta \sum_{i=1}^{3} B_{\tau-1}^{(i)} \left(X_{t+1}^{(i)} - \left(\frac{B_{\tau}^{(i)} - 1}{B_{\tau-1}^{(i)}}\right) X_t^{(i)} \right) + \Delta \sum_{i=1}^{3} X_t^{(i)} - \Delta(X_t^{(1)} + X_t^{(2)}) \\ &= \log A_{\tau-1} - \log A_{\tau} - \Delta \sum_{i=1}^{2} B_{\tau-1}^{(i)} \left(X_{t+1}^{(i)} - \left(\frac{B_{\tau}^{(i)} - 1}{B_{\tau-1}^{(i)}}\right) X_t^{(i)} \right) + \Delta X_t^{(3)} \\ &\quad - \Delta B_{\tau-1}^{(3)} \left(X_{t+1}^{(3)} - \left(\frac{B_{\tau}^{(3)} - 1}{B_{\tau-1}^{(3)}}\right) X_t^{(3)} \right) \\ &= \log A_{\tau-1} - \log A_{\tau} - \Delta \sum_{i=1}^{2} B_{\tau-1}^{(i)} \left(X_{t+1}^{(i)} - \left(1 - \kappa_{i,i}^{\mathbb{Q}}\right) X_t^{(i)} \right) + \Delta X_t^{(3)} \\ &\quad - \Delta B_{\tau-1}^{(3)} \left(X_{t+1}^{(3)} - \left((1-\lambda) - \frac{(1-\lambda)^{\tau-1}}{B_{\tau-1}^{(3)}}\right) X_t^{(3)} \right) \\ &= \log A_{\tau-1} - \log A_{\tau} - \Delta \sum_{i=1}^{3} B_{\tau-1}^{(i)} \left(X_{t+1}^{(i)} - \left(1 - \kappa_{i,i}^{\mathbb{Q}}\right) X_t^{(i)} \right) + \Delta\left(1 - (1-\lambda)^{\tau-1}\right) X_t^{(3)}. \end{aligned} \tag{4.2}$$

Mimicking Augustyniak et al. (2021) by considering a portfolio consisting of M positions with weights $\omega_1, \ldots, \omega_M$ (where $\sum_{j=1}^{M} \omega_j = 1$) that are rolled-over on bonds with fixed time-to-maturities $\tau_1, \ldots, \tau_M$, the log-return of the rolling bond fund between times t and $t+1$, denoted by $R_{t+1}^{(B)}$, is approximately characterized by

$$
\begin{aligned}
R_{t+1}^{(B)} - \Delta r_t &\approx \omega_1 R_{t+1}^{(\tau_1)} + \cdots + \omega_M R_{t+1}^{(\tau_M)} - \Delta r_t \\
&= \omega_1 \left(R_{t+1}^{(\tau_1)} - \Delta r_t \right) + \cdots + \omega_M \left(R_{t+1}^{(\tau_M)} - \Delta r_t \right) \\
&= \sum_{j=1}^{M} \omega_j \Bigg(\log A_{\tau_j - 1} - \log A_{\tau_j} - \Delta \sum_{i=1}^{3} B_{\tau_j - 1}^{(i)} \left(X_{t+1}^{(i)} - \left(1 - \kappa_{i,i}^{\mathbb{Q}}\right) X_t^{(i)} \right) \\
&\quad + \Delta \left(1 - (1-\lambda)^{\tau_j - 1} \right) X_t^{(3)} \Bigg) \\
&= \sum_{j=1}^{M} \omega_j \left(\log A_{\tau_j - 1} - \log A_{\tau_j} \right) - \Delta \sum_{i=1}^{3} \sum_{j=1}^{M} \omega_j B_{\tau_j - 1}^{(i)} \left(X_{t+1}^{(i)} - \left(1 - \kappa_{i,i}^{\mathbb{Q}}\right) X_t^{(i)} \right) \\
&\quad + \Delta \sum_{j=1}^{M} \omega_j \left(1 - (1-\lambda)^{\tau_j - 1} \right) X_t^{(3)}.
\end{aligned}
\tag{4.3}
$$

Relationship (4.3) will serve as the basis to model the fixed income part of the mixed fund in the proposed model. The key difference with the Augustyniak et al. (2021) model is the presence of the last term proportional to $X_t^{(3)}$ in (4.3), which is not present in their work.

4.2 Risky assets returns

The adaptation of the mixed fund model proposed in this paper keeps the same model for excess equity returns as in Augustyniak et al. (2021). Returns of q equity assets are represented with the exponential generalized auto-regressive conditional heteroskedastic (EGARCH) model of Nelson (1991):

$$
R_{t+1,j}^{(S)} - \Delta r_t = \lambda_j^{(S)} \sqrt{h_{t,j}^{(S)}} - \frac{1}{2} h_{t,j}^{(S)} + \sqrt{h_{t,j}^{(S)}} Z_{t+1,j}^{(S)}, \tag{4.4}
$$

$$
\log h_{t,j}^{(S)} = \omega_j^{(S)} + \alpha_j^{(S)} Z_{t,j}^{(S)} + \gamma_j^{(S)} \left(|Z_{t,j}^{(S)}| - \frac{2}{\sqrt{2\pi}} \right) + \beta_j^{(S)} \log h_{t-1,j}^{(S)}, \tag{4.5}
$$

where $(\omega_j^{(S)}, \alpha_j^{(S)}, \gamma_j^{(S)}, \beta_j^{(S)})$ are parameters associated with the conditional volatility. The process $\{h_{t,j}^{(S)}\}_{t=1}^{T-1}$ is $\mathcal{F}$-adapted. Parameters $\lambda_j^{(S)}$ represent equity risk premia. For $j = 1, \ldots, q$, $Z_j^{(S)} \equiv \{Z_{t,j}^{(S)}\}_{t=1}^{T}$ are standard Gaussian white noises under $\mathbb{P}$, independent of Z_t, where the correlation between $Z_{t,i}^{(S)}$ and $Z_{t,j}^{(S)}$ is denoted $\Gamma_{i,j}$.

As in Augustyniak et al. (2021), the following risk-neutral dynamics for the q equity assets is assumed:

$$R_{t+1,j}^{(S)} - \Delta r_t = -\frac{1}{2}h_{t,j}^{(S)} + \sqrt{h_{t,j}^{(S)}}Z_{t+1,j}^{\mathbb{Q}(S)}, \tag{4.6}$$

$$\log h_{t,j}^{(S)} = \omega_j^{(S)} + \alpha_j^{(S)}(Z_{t,j}^{\mathbb{Q}(S)} - \lambda_j^{(S)}) + \gamma_j^{(S)}\left(|Z_{t,j}^{\mathbb{Q}(S)} - \lambda_j^{(S)}| - \frac{2}{\sqrt{2\pi}}\right) + \beta_j^{(S)}\log h_{t-1,j}^{(S)}, \tag{4.7}$$

where, under the risk-neutral measure $\mathbb{Q}$, $\{Z_{t,j}^{\mathbb{Q}(S)}\}_{t=1}^T = \{Z_{t,j}^{(S)} + \lambda_j^{(S)}\}_{t=1}^T$ are also standard Gaussian white noises for each $j = 1, \dots, q$, still with a contemporaneous correlation matrix $\Gamma = [\Gamma_{i,j}]_{i,j=1}^q$. The independence between the equity innovations $\{Z_{t,j}^{\mathbb{Q}(S)}\}_{t=1}^T$ and interest rate innovations $\{Z_t^{\mathbb{Q}}\}_{t=1}^T$ is assumed to also hold under $\mathbb{Q}$.

4.3 Mixed fund dynamics

To motivate the structure of the proposed mixed fund model, consider a portfolio invested in the rolling bond fund with weight W and in equity indices with respective weights $\tilde{\psi}_1^{(S)}, \dots, \tilde{\psi}_q^{(S)}$, where $W + \sum_{j=1}^q \tilde{\psi}_j^{(S)} = 1$. Its excess return would be represented as

$$\begin{aligned}\tilde{R}_{t+1}^{(F)} - \Delta r_t &\approx W(R_{t+1}^{(B)} - \Delta r_t) + \sum_{j=1}^q \tilde{\psi}_j^{(S)}\left(R_{t+1,j}^{(S)} - \Delta r_t\right) \\ &= \tilde{\psi}_0 + \sum_{i=1}^3 \tilde{\psi}_i\left(X_{t+1}^{(i)} - \left(1 - \kappa_{i,i}^{\mathbb{Q}}\right)X_t^{(i)}\right) + \tilde{\psi}_3' X_t^{(3)} + \sum_{j=1}^q \tilde{\psi}_j^{(S)}\left(R_{t+1,j}^{(S)} - \Delta r_t\right)\end{aligned} \tag{4.8}$$

where

$$\tilde{\psi}_0 = W\sum_{j=1}^M \omega_j\left(\log A_{\tau_j-1} - \log A_{\tau_j}\right), \tag{4.9}$$

$$\tilde{\psi}_i = -W\Delta\sum_{j=1}^M \omega_j B_{\tau_j-1}^{(i)}, \quad i = 1, 2, 3, \tag{4.10}$$

$$\tilde{\psi}_3' = W\Delta\sum_{j=1}^M \omega_j\left(1 - (1-\lambda)^{\tau_j-1}\right). \tag{4.11}$$

Equation (4.8) serves as the conceptual basis for the proposed mixed fund model. However, instead of relying on (4.9)-(4.11), model parameters can be directly estimated from the data in an econometric fashion. Indeed, allocations to fixed income and equity assets from the mixed funds are not exactly identical to those implied by previous assumptions–that is an investment in a rolling horizon bond fund and in equity indices–and basis risk needs to be taken into account. Moreover, as in Augustyniak et al. (2021), basis risk is further represented through a noise component also following EGARCH dynamics. Mixed fund excess returns are thus assumed to be of the form

$$R_{t+1}^{(F)} - \Delta r_t = \psi_0 + \sum_{i=1}^{3} \psi_i \left(X_{t+1}^{(i)} - \left(1 - \kappa_{i,i}^{\mathbb{Q}}\right) X_t^{(i)} \right) + \psi_3' X_t^{(3)} + \sum_{j=1}^{q} \psi_j^{(S)} \left(R_{t+1,j}^{(S)} - \Delta r_t \right) + \sqrt{h_t^{(F)}} Z_{t+1}^{(F)}, \tag{4.12}$$

$$\log h_t^{(F)} = \omega^{(F)} + \alpha^{(F)} Z_t^{(F)} + \gamma^{(F)} \left(|Z_t^{(F)}| - \frac{2}{\sqrt{2\pi}} \right) + \beta^{(F)} \log h_{t-1}^{(F)}, \tag{4.13}$$

where the basis risk innovation process $Z^{(F)} = \{Z_t^{(F)}\}_{t=1}^T$ is a standard Gaussian white noise under $\mathbb{P}$, independent of interest rate innovations $Z^{\mathbb{P}}$ and equity innovations $Z_j^{(S)}, j = 1, \ldots, q$. Basis risk volatility parameters $(\omega^{(F)}, \alpha^{(F)}, \gamma^{(F)}, \beta^{(F)})$ and linear coefficients $(\psi_0, \psi_1, \psi_2, \psi_3, \psi_3', \psi_1^{(S)}, \ldots, \psi_q^{(S)})$ are model parameters to be estimated.

The risk-neutral dynamics can also be obtained along the lines of Augustyniak et al. (2021) by translating basis risk innovations $\{Z_t^{(F)}\}_{t=1}^T$ by some adapted process $\{\lambda_t^{(F)}\}_{t=0}^{T-1}$: under $\mathbb{Q}$, innovations $\{Z_t^{\mathbb{Q}(F)}\}_{t=1}^T \equiv \{Z_t^{(F)} + \lambda_{t-1}^{(F)}\}_{t=1}^T$ form a Gaussian white noise independent of risk-neutral term structure and equity innovations $\{Z_t^{\mathbb{Q}}\}_{t=1}^T$ and $\{Z_{t,j}^{\mathbb{Q}(S)}\}_{t=1}^T$, respectively. The determination of the process $\{\lambda_t^{(F)}\}_{t=0}^{T-1}$ stems from the martingale property that must be satisfied by the mixed fund value process. Its value, which characterizes the risk-neutral dynamics of the mixed fund, is provided in the following proposition whose proof is in Appendix A.

Proposition 4.1. *The risk-neutral dynamics of mixed fund excess returns are given by*

$$R_{t+1}^{(F)} - \Delta r_t = -\frac{1}{2} \left(\sigma_t^{(F)} \right)^2 + \sigma_t^{(F)} \epsilon_{t+1}^{\mathbb{Q}(F)} \tag{4.14}$$

where

$$
\begin{aligned}
\left(\sigma_t^{(F)}\right)^2 &\equiv \sum_{i=1}^{3}\sum_{l=1}^{3}\psi_i\psi_l\Sigma_{i,i}\Sigma_{l,l}\rho_{i,l} + \sum_{j=1}^{q}\sum_{k=1}^{q}\psi_j^{(S)}\psi_k^{(S)}\Gamma_{j,k}\sqrt{h_{t,j}^{(S)}h_{t,k}^{(S)}} + h_t^{(F)}, \qquad (4.15)\\
\phi_t &\equiv \psi_0 + \sum_{i=1}^{3}\psi_i\left(\kappa_{i,i}^{\mathbb{Q}}\theta_i^{\mathbb{Q}} - \lambda\theta_3^{\mathbb{Q}}\mathbb{1}_{\{i=2\}}\right) + (\psi_2\lambda + \psi_3')X_t^{(3)} - \sum_{j=1}^{q}\psi_j^{(S)}\left(\frac{1}{2}h_{t,j}^{(S)}\right),\\
\epsilon_{t+1}^{\mathbb{Q}(F)} &\equiv \frac{\sum_{i=1}^{p}\psi_i\Sigma_{i,i}Z_{t+1}^{\mathbb{Q}(i)} + \sum_{j=1}^{q}\psi_j^{(S)}\sqrt{h_{t,j}^{(S)}}Z_{t+1,j}^{\mathbb{Q}(S)} + \sqrt{h_t^{(F)}}Z_{t+1}^{\mathbb{Q}(F)}}{\sigma_t^{(F)}},\\
\log h_t^{(F)} &= \omega^{(F)} + \alpha^{(F)}(Z_t^{\mathbb{Q}(F)} - \lambda_{t-1}^{(F)}) + \gamma^{(F)}\left(|Z_t^{\mathbb{Q}(F)} - \lambda_{t-1}^{(F)}| - \frac{2}{\sqrt{2\pi}}\right) + \beta^{(F)}\log h_{t-1}^{(F)},\\
\lambda_t^{(F)} &\equiv \frac{1}{\sqrt{h_t^{(F)}}}\left[\phi_t + \frac{1}{2}\left(\sigma_t^{(F)}\right)^2\right].
\end{aligned}
$$

and the process $\{\epsilon_{t+1}^{\mathbb{Q}(F)}\}_{t=1}^{T}$ *is a sequence of independent standardized Gaussian variables under* $\mathbb{Q}$.

4.4 Estimation of the mixed fund model

The mixed fund model developed above is estimated on real data. The equity model (4.4) and (4.5) is estimated for two equity assets ($q = 2$), namely the *S&P/TSX Composite* and *S&P 500* stock indices. For the mixed fund model (4.12)-(4.13), we consider the *Assumption/Louisbourg Balanced Fund A*.[3] This is a mixed bond and equity fund composed approximately of 39% of Canadian fixed income, 36% of Canadian equity, 15% of US equity, and 10% of other products. Monthly price return data (NAV returns for the mixed fund) in Canadian currency from February 1986 (from February 1996 for the mixed fund) to January 2022 are considered, a span that matches the term structure data used in earlier sections. Note that for the mixed fund, 14 out of the 326 monthly returns are missing values.

Table 4 and Table 5 respectively show maximum likelihood estimated parameter and corresponding standard errors for the equity model (4.4)-(4.5) and the mixed fund model (4.12)-(4.13).[4] Note that initial variances $h_{0,j}^{(S)}$, $j = 1, 2$ and $h_0^{(F)}$ are also considered as parameters to optimize during the estimation. Equity indices are highly correlated with $\Gamma_{1,2} = 0.753$. Furthermore, the

[3] https://assumption.lipperweb.com/assumplife/list#FundDetail

[4] Monthly periods with a missing value for the mixed fund return are discarded in the likelihood calculation.

persistence of the volatility for both indices is high and similar, with $\beta_1^{(S)} = 0.628$ for the S&P TSX Composite and $\beta_2^{(S)} = 0.697$ the S&P 500. Furthermore, the presence of a leverage effect is implied by negative values of $\alpha_j^{(S)}$, $j = 1, 2$. For the mixed fund model, all parameter estimates are significant at the 99% confidence level. Negative values for estimates of coefficient ψ_i, $i = 1, 2$ reflect negative correlation between variations of the short rate and returns of the mixed fund. This is expected due to the fund being partially invested in fixed income, where bond prices are inversely related to interest rates. Interestingly, the estimate of coefficient ψ_3', a parameter not present in the Augustyniak et al. (2021) model, is also significant and negative. It implies that mixed fund returns are negatively related to the humpiness of the term structure (and not only to changes in humpiness).

Table 4: Bivariate EGARCH equity model parameter estimates

Stock index	j	λ_j^S	$\omega_j^{(S)}$	$\alpha_j^{(S)}$	$\gamma_j^{(S)}$	$\beta_j^{(S)}$	$\Gamma_{1,2}$	$\sqrt{12h_{0,j}^{(S)}}$
S&P/TSX	1	0.08443	-2.38375	-0.16171	0.38711	0.62836		14.69%
		(0.04574)	(0.53917)	(0.06079)	(0.09043)	(0.08330)	0.75281	
S&P 500	2	0.12605	-1.92871	-0.14922	0.32486	0.69715	(0.02126)	14.98%
		(0.04390)	(0.61691)	(0.05631)	(0.08286)	(0.09706)		

Notes: Maximum likelihood estimates and standard errors (in parentheses) of the equity model (4.4)-(4.5). Indices $j = 1$ and $j = 2$ are respectively the S&P/TSX Composite and the S&P 500. The estimation is performed on monthly price return time series extending from February 1986 to January 2022 (432 returns by index).

Table 5: Mixed fund model parameter estimates

Parameter	ψ_0	ψ_1	ψ_2	ψ_3	ψ_3'	$\psi_1^{(S)}$
Estimate	0.00404	-1.72446	-0.76950	-0.44537	-0.05524	0.40076
	(0.00123)	(0.40433)	(0.24252)	(0.11151)	(0.01935)	(0.01464)
Parameter	$\psi_2^{(S)}$	$\omega^{(F)}$	$\alpha^{(F)}$	$\gamma^{(F)}$	$\beta^{(F)}$	$\sqrt{12h_0^{(F)}}$
Estimate	0.13458	-1.03122	-0.09390	0.06653	0.89453	2.61%
	(0.01390)	(0.01238)	(0.03527)	(0.02091)	(0.00028)	

Notes: Maximum likelihood estimates and standard errors (in parentheses) of the mixed fund model (4.12)-(4.13). The estimation is performed on a monthly NAV return time series for the Assumption/Louisbourg Balanced Fund A extending from February 1996 to January 2022 (312 returns).

5 Conclusion

This paper develops a discrete-time version of the arbitrage-free Nelson-Siegel model for dynamic term structure modeling which is slightly different than but similar to that in Hong et al. (2016). A closed-form solution for the price of risk-free zero-coupon bonds is obtained, which makes it possible to devise a convenient Kalman filter-based joint estimation procedure for physical and risk-neutral dynamics of factors driving the term structure. The estimation of the model on historical data for the Canadian spot curve reveal that the model accurately represents term structure movements and possesses superior distributional predictive power in comparison to some version of the dynamic Nelson-Siegel model and to a discrete-time G3 model previously proposed by Augustyniak et al. (2021), at least for the considered dataset. The discrete-time arbitrage-free Nelson-Siegel model also has two other advantages over the latter benchmark: it tends to produce less frequent negative short rates and it provides better interpretability for the three factors underlying the model. Finally, the mixed fund model of Augustyniak et al. (2021) characterizing the dynamics of a mutual fund invested in both equity and fixed income is adapted to integrate the discrete-time arbitrage-free Nelson-Siegel model as its component to depict interest rates dynamics.

A Proofs

In this section, proofs to several results mentioned in the body of the manuscript are provided.

To prove Proposition 2.1, the following lemma is needed.

Lemma A.1. *Assume (2.10) holds. For $t = 0, \dots, T$ and $n = 0, \dots, T - t$,*

$$\begin{aligned} X^{(i)}_{t+n} =& X^{(i)}_t (1-\kappa^{\mathbb{Q}}_{i,i})^n + \kappa^{\mathbb{Q}}_{i,i}\theta^{\mathbb{Q}}_i \sum_{l=1}^{n} (1-\kappa^{\mathbb{Q}}_{i,i})^{(n-l)} + \Sigma_{i,i} \sum_{l=1}^{n} Z^{\mathbb{Q}}_{t+l,i} (1-\kappa^{\mathbb{Q}}_{i,i})^{(n-l)} \\ &+ \sum_{l=1}^{n} \sum_{j\neq i}^{3} \kappa^{\mathbb{Q}}_{i,j} (\theta^{\mathbb{Q}}_j - X^{(j)}_{t+l-1})(1-\kappa^{\mathbb{Q}}_{i,i})^{(n-l)}. \end{aligned} \tag{A.1}$$

Proof of Lemma A.1: The proof relies on induction. The case $n = 0$ is trivial when using the convention $\sum_{l=1}^{0} x_l \equiv 0$ for any sequence of real numbers $\{x_l\}$.

Then, assume (A.1) holds for some $n < T - t - 1$. Using (2.10), for any $i = 1, 2, 3$,

$$\begin{aligned} X_{t+n+1}^{(i)} =& X_{t+n}^{(i)} + \sum_{j=1}^{3} \kappa_{i,j}^{\mathbb{Q}} (\theta_j^{\mathbb{Q}} - X_{t+n}^{(j)}) + \Sigma_{i,i} Z_{t+n+1,i}^{\mathbb{Q}} \\ =& X_{t+n}^{(i)} (1 - \kappa_{i,i}^{\mathbb{Q}}) + \kappa_{i,i}^{\mathbb{Q}} \theta_i^{\mathbb{Q}} + \sum_{j \neq i}^{3} \kappa_{i,j}^{\mathbb{Q}} (\theta_j^{\mathbb{Q}} - X_{t+n}^{(j)}) + \Sigma_{i,i} Z_{t+n+1,i}^{\mathbb{Q}}. \end{aligned}$$

Applying (A.1) in the latter equality yields

$$\begin{aligned} X_{t+n+1}^{(i)} =& X_t^{(i)} (1 - \kappa_{i,i}^{\mathbb{Q}})^{n+1} + \kappa_{i,i}^{\mathbb{Q}} \theta_i^{\mathbb{Q}} \sum_{l=1}^{n} (1 - \kappa_{i,i}^{\mathbb{Q}})^{(n+1-l)} + \Sigma_{i,i} \sum_{l=1}^{n} Z_{t+l,i}^{\mathbb{Q}} (1 - \kappa_{i,i}^{\mathbb{Q}})^{(n+1-l)} \\ &+ \sum_{l=1}^{n} \sum_{j \neq i}^{3} \kappa_{i,j}^{\mathbb{Q}} (\theta_j^{\mathbb{Q}} - X_{t+l-1}^{(j)}) (1 - \kappa_{i,i}^{\mathbb{Q}})^{(n+1-l)} + \kappa_{i,i}^{\mathbb{Q}} \theta_i^{\mathbb{Q}} + \sum_{j \neq 1}^{3} \kappa_{i,j}^{\mathbb{Q}} (\theta_j^{\mathbb{Q}} - X_{t+n}^{(j)}) + \Sigma_{i,i} Z_{t+n+1,i}^{\mathbb{Q}} \\ =& X_t^{(i)} (1 - \kappa_{i,i}^{\mathbb{Q}})^{n+1} + \kappa_{i,i}^{\mathbb{Q}} \theta_i^{\mathbb{Q}} \sum_{l=1}^{n+1} (1 - \kappa_{i,i}^{\mathbb{Q}})^{(n+1-l)} + \Sigma_{i,i} \sum_{l=1}^{n+1} Z_{t+l,i}^{\mathbb{Q}} (1 - \kappa_{i,i}^{\mathbb{Q}})^{(n+1-l)} \\ &+ \sum_{l=1}^{n+1} \sum_{j \neq i}^{3} \kappa_{i,j}^{\mathbb{Q}} (\theta_j^{\mathbb{Q}} - X_{t+l-1}^{(j)}) (1 - \kappa_{i,i}^{\mathbb{Q}})^{(n+1-l)}, \end{aligned}$$

hence completing the induction. $\square$

Remark A.1. *The result of Lemma A.1 is also valid under the $\mathbb{P}$-measure, i.e. when replacing all $\mathbb{Q}$'s with $\mathbb{P}$'s in the statement of the lemma and the proof.*

Proof of Lemma 2.1: Equation (2.11) is trivial. To show (2.12),

$$\begin{aligned} \sum_{u=1}^{\tau-1} u r^u &= \sum_{u=1}^{\tau-1} r \frac{d}{dr} r^u \\ &= r \frac{d}{dr} \sum_{u=1}^{\tau-1} r^u \\ &= r \frac{(1 - \tau r^{\tau-1})(1 - r) + (r - r^{\tau})}{(1 - r)^2} \\ &= r \frac{1 - \tau r^{\tau-1} + \tau r^{\tau} - r^{\tau}}{(1 - r)^2} \\ &= \frac{r - \tau r^{\tau} + (\tau - 1) r^{\tau+1}}{(1 - r)^2}. \end{aligned}$$

Finally, to show (2.13),

$$\begin{aligned}
\sum_{u=1}^{\tau-1} u^2 r^u &= r\sum_{u=1}^{\tau-1} u\frac{d}{dr}r^u \\
&= r\frac{d}{dr}\sum_{u=1}^{\tau-1} ur^u \\
&= r\frac{d}{dr}\frac{r-\tau r^\tau+(\tau-1)r^{\tau+1}}{(1-r)^2} \\
&= r\frac{(1-r)^2[1-\tau^2r^{\tau-1}+(\tau+1)(\tau-1)r^\tau]+2(1-r)[r-\tau r^\tau+(\tau-1)r^{\tau+1}]}{(1-r)^4} \\
&= \frac{[r-r^2-\tau^2r^\tau+\tau^2r^{\tau+1}+(\tau+1)(\tau-1)r^{\tau+1}-(\tau+1)(\tau-1)r^{\tau+2}]+2[r^2-\tau r^{\tau+1}+(\tau-1)r^{\tau+2}]}{(1-r)^3} \\
&= \frac{-(\tau-1)^2r^{\tau+2}+(2\tau^2-2\tau-1)r^{\tau+1}-\tau^2r^\tau+r^2+r}{(1-r)^3}. \quad \square
\end{aligned}$$

Proof of Proposition 2.1: From (2.7), for any $\tau = 1, \dots, T-t$,

$$\sum_{s=0}^{\tau-1} r_{t+s} = \sum_{s=0}^{\tau-1} X_{t+s}^{(1)} + \sum_{s=0}^{\tau-1} X_{t+s}^{(2)}.$$

Furthermore, since $\kappa_{1,j}^{\mathbb{Q}} = 0$ for $j = 1, \dots, 3$, Equation (A.1) leads to

$$\begin{aligned}
\sum_{s=0}^{\tau-1} X_{t+s}^{(1)} &= \sum_{s=0}^{\tau-1}\left[X_t^{(1)} + \Sigma_{1,1}\sum_{l=1}^{s} Z_{t+l,1}^{\mathbb{Q}}\right] \\
&= \tau X_t^{(1)} + \Sigma_{1,1}\sum_{s=1}^{\tau-1}\sum_{l=1}^{\tau-1} \mathbb{1}_{\{l\leq s\}} Z_{t+l,1}^{\mathbb{Q}} \\
&= \tau X_t^{(1)} + \Sigma_{1,1}\sum_{l=1}^{\tau-1}\sum_{s=l}^{\tau-1} Z_{t+l,1}^{\mathbb{Q}} \\
&= \tau X_t^{(1)} + \Sigma_{1,1}\sum_{l=1}^{\tau-1}(\tau-l) Z_{t+l,1}^{\mathbb{Q}}. \qquad \text{(A.2)}
\end{aligned}$$

Moreover, relying again on Equation (A.1) and recalling that $\kappa^{\mathbb{Q}}_{2,1} = 0$, $\kappa^{\mathbb{Q}}_{2,2} = \lambda$ and $\kappa^{\mathbb{Q}}_{2,3} = -\lambda$,

$$\begin{aligned}
\sum_{s=0}^{\tau-1} X^{(2)}_{t+s} = &\sum_{s=0}^{\tau-1} \Bigg\{ X^{(2)}_t (1-\lambda)^s + \lambda\theta^{\mathbb{Q}}_2 \sum_{l=1}^{s} (1-\lambda)^{s-l} + \Sigma_{2,2} \sum_{l=1}^{s} Z^{\mathbb{Q}}_{t+l,2} (1-\lambda)^{s-l} \\
&- \lambda \sum_{l=1}^{s} (1-\lambda)^{s-l} \theta^{\mathbb{Q}}_3 + \lambda \sum_{l=1}^{s} (1-\lambda)^{s-l} X^{(3)}_{t+l-1} \Bigg\} \\
=& X^{(2)}_t \sum_{s=0}^{\tau-1} (1-\lambda)^s + \lambda\theta^{\mathbb{Q}}_2 \sum_{s=1}^{\tau-1}\sum_{l=1}^{s} (1-\lambda)^{s-l} + \Sigma_{2,2} \sum_{s=1}^{\tau-1}\sum_{l=1}^{s} (1-\lambda)^{s-l} Z^{\mathbb{Q}}_{t+l,2} \\
&- \lambda\theta^{\mathbb{Q}}_3 \sum_{s=1}^{\tau-1}\sum_{l=1}^{s} (1-\lambda)^{s-l} + \lambda \sum_{s=1}^{\tau-1}\sum_{l=1}^{s} (1-\lambda)^{s-l} X^{(3)}_{t+l-1} \\
=& X^{(2)}_t \sum_{s=0}^{\tau-1} (1-\lambda)^s + \lambda \left(\theta^{\mathbb{Q}}_2 - \theta^{\mathbb{Q}}_3\right) \sum_{s=1}^{\tau-1}\sum_{l=0}^{s-1} (1-\lambda)^l + \sum_{s=1}^{\tau-1}\sum_{l=1}^{\tau-1} \mathbb{1}_{\{l\le s\}} (1-\lambda)^{s-l} \left(\Sigma_{2,2} Z^{\mathbb{Q}}_{t+l,2} + \lambda X^{(3)}_{t+l-1}\right) \\
=& X^{(2)}_t \sum_{s=0}^{\tau-1} (1-\lambda)^s + \lambda \left(\theta^{\mathbb{Q}}_2 - \theta^{\mathbb{Q}}_3\right) \sum_{s=1}^{\tau-1} \frac{1-(1-\lambda)^s}{\lambda} + \sum_{l=1}^{\tau-1}\sum_{s=1}^{\tau-1} \mathbb{1}_{\{l\le s\}} (1-\lambda)^{s-l} \left(\Sigma_{2,2} Z^{\mathbb{Q}}_{t+l,2} + \lambda X^{(3)}_{t+l-1}\right) \\
=& X^{(2)}_t \frac{1-(1-\lambda)^\tau}{\lambda} + \left(\theta^{\mathbb{Q}}_2 - \theta^{\mathbb{Q}}_3\right) \left(\tau - 1 - \frac{1-\lambda-(1-\lambda)^\tau}{\lambda}\right) \\
&+ \sum_{l=1}^{\tau-1} \left(\Sigma_{2,2} Z^{\mathbb{Q}}_{t+l,2} + \lambda X^{(3)}_{t+l-1}\right) \sum_{s=1}^{\tau-1} \mathbb{1}_{\{l\le s\}} (1-\lambda)^{s-l}.
\end{aligned} \tag{A.3}$$

Note that

$$\sum_{s=1}^{\tau-1} \mathbb{1}_{\{l\le s\}} (1-\lambda)^{s-l} = \sum_{s=l}^{\tau-1} (1-\lambda)^{s-l} = \sum_{s=0}^{\tau-1-l} (1-\lambda)^s = \frac{1-(1-\lambda)^{\tau-l}}{\lambda}. \tag{A.4}$$

Placing (A.4) in (A.3) yields

$$\begin{aligned}
\sum_{s=0}^{\tau-1} X^{(2)}_{t+s} =& X^{(2)}_t \frac{1-(1-\lambda)^\tau}{\lambda} + \left(\theta^{\mathbb{Q}}_2 - \theta^{\mathbb{Q}}_3\right) \left(\tau - 1 - \frac{1-\lambda-(1-\lambda)^\tau}{\lambda}\right) \\
&+ \Sigma_{2,2} \sum_{l=1}^{\tau-1} Z^{\mathbb{Q}}_{t+l,2} \frac{1-(1-\lambda)^{\tau-l}}{\lambda} + \sum_{l=1}^{\tau-1} \left[1-(1-\lambda)^{\tau-l}\right] X^{(3)}_{t+l-1}.
\end{aligned} \tag{A.5}$$

Additionally, combining (A.1) with $\kappa^{\mathbb{Q}}_{3,1} = \kappa^{\mathbb{Q}}_{3,2} = 0$ and $\kappa^{\mathbb{Q}}_{3,3} = \lambda$ leads to

$$
\begin{aligned}
&\sum_{l=1}^{\tau-1}\left[1-(1-\lambda)^{\tau-l}\right]X_{t+l-1}^{(3)}\\
&=\sum_{l=0}^{\tau-2}\left[1-(1-\lambda)^{\tau-l-1}\right]X_{t+l}^{(3)}\\
&=\sum_{l=0}^{\tau-2}\left[1-(1-\lambda)^{\tau-l-1}\right]\left[X_t^{(3)}(1-\lambda)^l+\lambda\theta_3^{\mathbb{Q}}\sum_{k=1}^{l}(1-\lambda)^{(l-k)}+\Sigma_{3,3}\sum_{k=1}^{l}Z_{t+k,3}^{\mathbb{Q}}(1-\lambda)^{(l-k)}\right]\\
&=X_t^{(3)}\left(\sum_{l=0}^{\tau-2}(1-\lambda)^l-\sum_{l=0}^{\tau-2}(1-\lambda)^{\tau-1}\right)+\lambda\theta_3^{\mathbb{Q}}\sum_{l=0}^{\tau-2}\frac{1-(1-\lambda)^l-(1-\lambda)^{\tau-l-1}+(1-\lambda)^{\tau-1}}{\lambda}\\
&+\Sigma_{3,3}\sum_{l=0}^{\tau-2}\sum_{k=1}^{\tau-2}\mathbb{1}_{\{k\le l\}}Z_{t+k,3}^{\mathbb{Q}}\left[(1-\lambda)^{(l-k)}-(1-\lambda)^{\tau-k-1}\right]\\
&=X_t^{(3)}\left(\frac{1-(1-\lambda)^{\tau-1}}{\lambda}-(\tau-1)(1-\lambda)^{\tau-1}\right)+\theta_3^{\mathbb{Q}}\sum_{l=0}^{\tau-2}\left(1-(1-\lambda)^l-(1-\lambda)^{\tau-l-1}+(1-\lambda)^{\tau-1}\right)\\
&\quad+\Sigma_{3,3}\sum_{k=1}^{\tau-2}Z_{t+k,3}^{\mathbb{Q}}\sum_{l=k}^{\tau-2}\left[(1-\lambda)^{(l-k)}-(1-\lambda)^{\tau-k-1}\right]\\
&=X_t^{(3)}\left(\frac{1-(1-\lambda)^{\tau-1}}{\lambda}-(\tau-1)(1-\lambda)^{\tau-1}\right)+\theta_3^{\mathbb{Q}}\Bigg((\tau-1)\left(1+(1-\lambda)^{\tau-1}\right)\\
&\quad-\frac{2-(1-\lambda)^{\tau-1}-\lambda-(1-\lambda)^{\tau}}{\lambda}\Bigg)+\Sigma_{3,3}\sum_{k=1}^{\tau-2}Z_{t+k,3}^{\mathbb{Q}}\left[\frac{1-(1-\lambda)^{(\tau-1-k)}}{\lambda}-(1-\lambda)^{\tau-k-1}(\tau-1-k)\right].
\end{aligned}
\tag{A.6}
$$

Placing (A.6) in (A.5) given

$$
\begin{aligned}
\sum_{s=0}^{\tau-1}X_{t+s}^{(2)}=&X_t^{(2)}\frac{1-(1-\lambda)^{\tau}}{\lambda}+X_t^{(3)}\left(\frac{1-(1-\lambda)^{\tau-1}}{\lambda}-(\tau-1)(1-\lambda)^{\tau-1}\right)\\
&+\theta_2^{\mathbb{Q}}\left(\tau-1-\frac{1-\lambda-(1-\lambda)^{\tau}}{\lambda}\right)+\theta_3^{\mathbb{Q}}\left((\tau-1)(1-\lambda)^{\tau-1}-\frac{1-(1-\lambda)^{\tau-1}}{\lambda}\right)\\
&+\Sigma_{2,2}\sum_{l=1}^{\tau-1}Z_{t+l,2}^{\mathbb{Q}}\frac{1-(1-\lambda)^{\tau-l}}{\lambda}+\Sigma_{3,3}\sum_{k=1}^{\tau-2}Z_{t+k,3}^{\mathbb{Q}}\left[\frac{1-(1-\lambda)^{(\tau-1-k)}}{\lambda}-(1-\lambda)^{\tau-k-1}(\tau-1-k)\right].
\end{aligned}
\tag{A.7}
$$

Define $R_{t,T} \equiv \Delta \sum_{j=t}^{T-1} r_j = \Delta \sum_{j=0}^{T-1} X_{t+j}^{(1)} + X_{t+j}^{(2)}$. Define also

$$B_\tau^{(1)} = \tau, \quad B_\tau^{(2)} = \frac{1-(1-\lambda)^\tau}{\lambda}, \quad B_\tau^{(3)} = \frac{1-(1-\lambda)^{\tau-1}}{\lambda} - (\tau-1)(1-\lambda)^{\tau-1}.$$

Based on (A.2) and (A.7), given $\mathcal{F}_t$, $-R_{t,T}$ has a Gaussian distribution with mean

$$-\Delta \sum_{j=1}^{3} B_\tau^{(j)} X_t^{(j)} - \Delta\theta_2^{\mathbb{Q}} \left(\tau - 1 - \frac{1-\lambda-(1-\lambda)^\tau}{\lambda}\right) - \Delta\theta_3^{\mathbb{Q}} \left((\tau-1)(1-\lambda)^{\tau-1} - \frac{1-(1-\lambda)^{\tau-1}}{\lambda}\right)$$

and variance $\Delta^2 \upsilon_\tau$ where

$$\upsilon_\tau = \sum_{i=1}^{3}\sum_{j=1}^{3} \upsilon_\tau^{(i,j)}$$

with

$$\begin{aligned}
\upsilon_\tau^{(1,1)} &= \mathrm{Var}^{\mathbb{Q}}\left(\Sigma_{1,1}\sum_{l=1}^{\tau-1}(\tau-l)Z_{t+l,1}^{\mathbb{Q}}\right) = \Sigma_{1,1}^2 \sum_{l=1}^{\tau-1}(\tau-l)^2 = \Sigma_{1,1}^2 \sum_{l=1}^{\tau-1} l^2 = \Sigma_{1,1}^2 \frac{\tau(\tau-1)(2\tau-1)}{6}, \\
\upsilon_\tau^{(2,2)} &= \mathrm{Var}^{\mathbb{Q}}\left(\Sigma_{2,2}\sum_{l=1}^{\tau-1} Z_{t+l,2}^{\mathbb{Q}} \frac{1-(1-\lambda)^{\tau-l}}{\lambda}\right) \\
&= \frac{\Sigma_{2,2}^2}{\lambda^2}\sum_{l=1}^{\tau-1}\left(1-(1-\lambda)^l\right)^2 \\
&= \frac{\Sigma_{2,2}^2}{\lambda^2}\left(\tau - 1 - 2\left[\frac{1-(1-\lambda)^\tau}{\lambda} - 1\right] + \frac{1-(1-\lambda)^{2\tau}}{1-(1-\lambda)^2} - 1\right) \\
&= \frac{\Sigma_{2,2}^2}{\lambda^2}\left(\tau - 2\left[\frac{1-(1-\lambda)^\tau}{\lambda}\right] + \frac{1-(1-\lambda)^{2\tau}}{1-(1-\lambda)^2}\right),
\end{aligned}$$

$$
\begin{aligned}
\upsilon_\tau^{(3,3)} &= \mathrm{Var}^{\mathbb{Q}}\left(\Sigma_{3,3}\sum_{k=1}^{\tau-2} Z^{\mathbb{Q}}_{t+k,3}\left[\frac{1-(1-\lambda)^{(\tau-1-k)}}{\lambda}-(1-\lambda)^{\tau-k-1}(\tau-1-k)\right]\right)\\
&= \Sigma_{3,3}^2\sum_{k=1}^{\tau-2}\left[\frac{1-(1-\lambda)^{(\tau-1-k)}}{\lambda}-(1-\lambda)^{\tau-k-1}(\tau-1-k)\right]^2\\
&= \frac{\Sigma_{3,3}^2}{\lambda^2}\sum_{k=1}^{\tau-2}\left[1-(1-\lambda)^k-\lambda(1-\lambda)^k k\right]^2\\
&= \frac{\Sigma_{3,3}^2}{\lambda^2}\sum_{k=1}^{\tau-2}\left[1+(1-\lambda)^{2k}+\lambda^2(1-\lambda)^{2k}k^2-2(1-\lambda)^k-2k\lambda(1-\lambda)^k+2\lambda k(1-\lambda)^{2k}\right]\\
&= \frac{\Sigma_{3,3}^2}{\lambda^2}\Big[\tau-2+\zeta_0\left((1-\lambda)^2,\tau-1\right)+\lambda^2\zeta_2\left((1-\lambda)^2,\tau-1\right)\\
&\quad -2\zeta_0\left((1-\lambda),\tau-1\right)-2\lambda\zeta_1\left((1-\lambda),\tau-1\right)+2\lambda\zeta_1\left((1-\lambda)^2,\tau-1\right)\Big]
\end{aligned}
$$

and

$$
\begin{aligned}
\upsilon_\tau^{(1,2)}=\upsilon_\tau^{(2,1)} &= \mathrm{Cov}^{\mathbb{Q}}\left(\Sigma_{1,1}\sum_{l=1}^{\tau-1}(\tau-l)Z^{\mathbb{Q}}_{t+l,1},\Sigma_{2,2}\sum_{l=1}^{\tau-1}Z^{\mathbb{Q}}_{t+l,2}\frac{1-(1-\lambda)^{\tau-l}}{\lambda}\right)\\
&= \rho_{1,2}\Sigma_{1,1}\Sigma_{2,2}\sum_{l=1}^{\tau-1} l\frac{1-(1-\lambda)^l}{\lambda}\\
&= \rho_{1,2}\Sigma_{1,1}\Sigma_{2,2}\frac{1}{\lambda}\left(\frac{\tau(\tau-1)}{2}-\zeta_1((1-\lambda),\tau)\right),\\
\upsilon_\tau^{(1,3)}=\upsilon_\tau^{(3,1)} &= \mathrm{Cov}^{\mathbb{Q}}\left(\Sigma_{1,1}\sum_{k=1}^{\tau-1}(\tau-k)Z^{\mathbb{Q}}_{t+k,1},\Sigma_{3,3}\sum_{k=1}^{\tau-2}Z^{\mathbb{Q}}_{t+k,3}\left[\frac{1-(1-\lambda)^{(\tau-1-k)}}{\lambda}-(1-\lambda)^{\tau-k-1}(\tau-1-k)\right]\right)\\
&= \rho_{1,3}\Sigma_{1,1}\Sigma_{3,3}\sum_{k=1}^{\tau-2}(\tau-k)\left[\frac{1-(1-\lambda)^{(\tau-1-k)}}{\lambda}-(1-\lambda)^{\tau-k-1}(\tau-1-k)\right]\\
&= \rho_{1,3}\Sigma_{1,1}\Sigma_{3,3}\sum_{k=1}^{\tau-2}(k+1)\left[\frac{1-(1-\lambda)^k}{\lambda}-(1-\lambda)^k k\right]\\
&= \rho_{1,3}\Sigma_{1,1}\Sigma_{3,3}\frac{1}{\lambda}\sum_{k=1}^{\tau-2}(k+1)\left[1-(1+\lambda k)(1-\lambda)^k\right]\\
&= \rho_{1,3}\Sigma_{1,1}\Sigma_{3,3}\frac{1}{\lambda}\left(\frac{(\tau-1)\tau}{2}-1-\sum_{k=1}^{\tau-2}(1+(\lambda+1)k+\lambda k^2)(1-\lambda)^k\right)\\
&= \rho_{1,3}\Sigma_{1,1}\Sigma_{3,3}\frac{1}{\lambda}\Big[\frac{\tau(\tau-1)}{2}-1-\zeta_0\left((1-\lambda),\tau-1\right)-(\lambda+1)\zeta_1\left((1-\lambda),\tau-1\right)\\
&\quad -\lambda\zeta_2\left((1-\lambda),\tau-1\right)\Big],
\end{aligned}
$$

$$
\begin{aligned}
\upsilon_\tau^{(2,3)} = \upsilon_\tau^{(3,2)} &= \mathrm{Cov}^{\mathbb{Q}}\Bigg(\Sigma_{2,2}\sum_{l=1}^{\tau-1} Z^{\mathbb{Q}}_{t+l,2}\frac{1-(1-\lambda)^{\tau-l}}{\lambda}, \Sigma_{3,3}\sum_{k=1}^{\tau-2} Z^{\mathbb{Q}}_{t+k,3}\Bigg[\frac{1-(1-\lambda)^{(\tau-1-k)}}{\lambda} \\
&\quad -(1-\lambda)^{\tau-k-1}(\tau-1-k)\Bigg]\Bigg) \\
&= \rho_{2,3}\Sigma_{2,2}\Sigma_{3,3}\sum_{k=1}^{\tau-2}\left(\frac{1-(1-\lambda)^{\tau-k}}{\lambda}\right)\left[\frac{1-(1-\lambda)^{\tau-k-1}}{\lambda} - (1-\lambda)^{\tau-k-1}(\tau-k-1)\right] \\
&= \rho_{2,3}\Sigma_{2,2}\Sigma_{3,3}\sum_{k=1}^{\tau-2}\left(\frac{1-(1-\lambda)^{k+1}}{\lambda}\right)\left[\frac{1-(1-\lambda)^{k}}{\lambda} - (1-\lambda)^{k}k\right] \\
&= \rho_{2,3}\Sigma_{2,2}\Sigma_{3,3}\sum_{k=1}^{\tau-2}\left(\frac{1-(2-\lambda)(1-\lambda)^k+(1-\lambda)(1-\lambda)^{2k}}{\lambda^2} + \frac{-(1-\lambda)^k k+(1-\lambda)(1-\lambda)^{2k}k}{\lambda}\right) \\
&= \rho_{2,3}\Sigma_{2,2}\Sigma_{3,3}\Bigg(\frac{\tau-2-(2-\lambda)\zeta_0\left((1-\lambda),\tau-1\right)+(1-\lambda)\zeta_0\left((1-\lambda)^2,\tau-1\right)}{\lambda^2} \\
&\quad + \frac{-\zeta_1\left((1-\lambda),\tau-1\right)+(1-\lambda)\zeta_1\left((1-\lambda)^2,\tau-1\right)}{\lambda}\Bigg). \square
\end{aligned}
$$

Lemma A.2. *Consider assumptions of Proposition 2.1. Then, for $i = 1, 2, 3$,*

$$
\frac{B_\tau^{(i)}-1}{B_{\tau-1}^{(i)}} = \left(1-\kappa^{\mathbb{Q}}_{i,i}\right) - \mathbb{1}_{\{i=3\}}\frac{(1-\lambda)^{\tau-1}}{B_{\tau-1}^{(3)}} = 1-\lambda\mathbb{1}_{\{i>1\}} - \mathbb{1}_{\{i=3\}}\frac{(1-\lambda)^{\tau-1}}{B_{\tau-1}^{(3)}}.
$$

Proof of Lemma A.2: Based on Proposition 2.1,

$$
\frac{B_\tau^{(1)}-1}{B_{\tau-1}^{(1)}} = \frac{\tau-1}{\tau-1} = 1.
$$

Furthermore,

$$
\frac{B_\tau^{(2)}-1}{B_{\tau-1}^{(2)}} = \frac{\frac{1-(1-\lambda)^{\tau}}{\lambda}-1}{\frac{1-(1-\lambda)^{\tau-1}}{\lambda}} = \frac{(1-\lambda)\frac{1-(1-\lambda)^{\tau-1}}{\lambda}}{\frac{1-(1-\lambda)^{\tau-1}}{\lambda}} = 1-\lambda.
$$

Finally,

$$\begin{aligned}
B_\tau^{(3)} &= \frac{1-(1-\lambda)^{\tau-1}}{\lambda} - (\tau-1)(1-\lambda)^{\tau-1} \\
&= (1-\lambda)\left(\frac{1-(1-\lambda)^{\tau-2}}{\lambda} + \frac{1}{1-\lambda} - (\tau-2)(1-\lambda)^{\tau-2} - (1-\lambda)^{\tau-2}\right) \\
&= B_{\tau-1}^{(3)}(1-\lambda) + 1 - (1-\lambda)^{\tau-1}. \ \square
\end{aligned}$$

Proof of Proposition 4.1: Using (2.10), (2.14), (4.6) and (4.12),

$$\begin{aligned}
R_{t+1}^{(F)} - \Delta r_t &= \psi_0 + \sum_{i=1}^{3} \psi_i \left(X_{t+1}^{(i)} - \left(1-\kappa_{i,i}^{\mathbb{Q}}\right) X_t^{(i)}\right) + \psi_3' X_t^{(3)} + \sum_{j=1}^{q} \psi_j^{(S)} \left(R_{t+1,j}^{(S)} - \Delta r_t\right) + \sqrt{h_t^{(F)}} Z_{t+1}^{(F)} \\
&= \psi_0 + \sum_{i=1}^{3} \psi_i \left(\Sigma_{i,i} Z_{t+1}^{\mathbb{Q}(i)} + \lambda\theta_i^{\mathbb{Q}} \mathbb{1}_{\{i>1\}} - \lambda\theta_3^{\mathbb{Q}} \mathbb{1}_{\{i=2\}} + \lambda \mathbb{1}_{\{i=2\}} X_t^{(3)}\right) + \psi_3' X_t^{(3)} \\
&\quad + \sum_{j=1}^{q} \psi_j^{(S)} \left(R_{t+1,j}^{(S)} - \Delta r_t\right) + \sqrt{h_t^{(F)}} Z_{t+1}^{(F)} \\
&= \underbrace{\psi_0 + \sum_{i=1}^{3} \psi_i \left(\kappa_{i,i}^{\mathbb{Q}} \theta_i^{\mathbb{Q}} - \lambda\theta_3^{\mathbb{Q}} \mathbb{1}_{\{i=2\}}\right) + (\psi_2\lambda + \psi_3') X_t^{(3)} - \sum_{j=1}^{q} \psi_j^{(S)} \left(\frac{1}{2} h_{t,j}^{(S)}\right)}_{=\phi_t} - \sqrt{h_t^{(F)}} \lambda_t^{(F)} \\
&\quad + \sum_{i=1}^{3} \psi_i \Sigma_{i,i} Z_{t+1}^{\mathbb{Q}(i)} + \sum_{j=1}^{q} \psi_j^{(S)} \sqrt{h_{t,j}^{(S)}} Z_{t+1,j}^{\mathbb{Q}(S)} + \sqrt{h_t^{(F)}} Z_{t+1}^{\mathbb{Q}(F)}.
\end{aligned}$$

Conditionally upon $\mathcal{F}_t$, $R_{t+1}^{(F)} - \Delta r_t$ is therefore Gaussian with variance

$$\mathrm{Var}^{\mathbb{Q}}\left[R_{t+1}^{(F)} - \Delta r_t \middle| \mathcal{F}_t\right] = \left(\sigma_t^{(F)}\right)^2 \tag{A.8}$$

with $\sigma_t^{(F)}$ given by (4.15). Enforcing that the discounted mixed fund value is a $\mathbb{Q}$-martingale requires

$$\begin{aligned}
1 &= \mathbb{E}^{\mathbb{Q}}\left[\exp\left(R_{t+1}^{(F)} - \Delta r_t\right) \middle| \mathcal{F}_t\right] \\
&= \exp\left(\phi_t - \sqrt{h_t^{(F)}} \lambda_t^{(F)} + \frac{1}{2}\left(\sigma_t^{(F)}\right)^2\right)
\end{aligned}$$

which leads to

$$\lambda_t^{(F)} = \frac{1}{\sqrt{h_t^{(F)}}}\left[\phi_t + \frac{1}{2}\left(\sigma_t^{(F)}\right)^2\right].$$

Finally, due to (A.8), the innovation

$$\epsilon_{t+1}^{\mathbb{Q}(F)} = \frac{\sum_{i=1}^{3}\psi_i\Sigma_{i,i}Z_{t+1}^{\mathbb{Q}(i)} + \sum_{j=1}^{q}\psi_j^{(S)}\sqrt{h_{t,j}^{(S)}}Z_{t+1,j}^{\mathbb{Q}(S)} + \sqrt{h_t^{(F)}}Z_{t+1}^{\mathbb{Q}(F)}}{\sigma_t^{(F)}}$$

is standard Gaussian under $\mathbb{Q}$, and thus $\{\epsilon_{t+1}^{\mathbb{Q}(F)}\}_{t=1}^{T}$ is a standard Gaussian white noise. $\square$

B Benchmarks

In this paper, the discrete-time Gaussian three-factor model introduced by Augustyniak et al. (2021) and the dynamic Nelson-Siegel model from (Diebold and Li, 2006) are considered as benchmarks for the DTAFNS model. Such benchmarks are presented below.

B.1 Discrete-time Gaussian three-factor model

Augustyniak et al. (2021) assume that the physical measure $\mathbb{P}$ dynamics of the short rate are that of a discrete-time version of the three-factor Gaussian G3 model:

$$\begin{aligned} r_t =& X_t^{(1)} + X_t^{(2)} + X_t^{(3)}, \\ X_{t+1}^{(i)} =& X_t^{(i)} + \kappa_i(\mu_i - X_t^{(i)}) + \sigma_i Z_{t+1}^{(i)}, \qquad i = 1,2,3, \end{aligned} \tag{B.1}$$

where $(\kappa_i, \mu_i, \sigma_i)$ are the model parameters and $\{Z_t^{(1)}, Z_t^{(2)}, Z_t^{(3)}\}_{t=1}^{T}$ is a Gaussian white noise process with contemporaneous correlation matrix Γ.

The risk-neutral dynamics of the factors is obtained using a discrete-time version of the Girsanov theorem. The processes $Z_i^{\mathbb{Q}} = \{Z_{t,i}^{\mathbb{Q}}\}_{t=1}^{T}$, $i = 1,2,3$ defined through $Z_{t+1,i}^{\mathbb{Q}} = Z_{t+1,i} - \lambda_i X_t^{(i)}$, with $\lambda_i \in \mathbb{R}$, are standard Gaussian white noises under the risk-neutral measure $\mathbb{Q}$, still with contemporaneous correlation matrix Γ. Therefore,

$$X_{t+1}^{(i)} = X_t^{(i)} + \kappa_i^{\mathbb{Q}}(\mu_i^{\mathbb{Q}} - X_t) + \sigma_i Z_{t+1}^{\mathbb{Q}}, \quad i = 1,2,3 \tag{B.2}$$

where

$$\kappa_i^{\mathbb{Q}} = \kappa_i - \sigma_i \lambda_i, \qquad \mu_i^{\mathbb{Q}} = \frac{\kappa_i \mu_i}{\kappa_i - \sigma_i \lambda_i}.$$

Augustyniak et al. (2021) show that under (B.2) the time-t price of a risk-free zero coupon bond paying one dollar at the maturity time T is given by

$$P(t,T) = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\Delta \sum_{j=t}^{T-t-1} r_{t-j}\right)\right] = \exp\left(A_\tau - \Delta \sum_{i=1}^{3} B_\tau^{(i)} X_t^{(i)}\right),$$

with $\tau = T - t$ and

$$A_\tau = \frac{\Delta^2}{2} \mathbb{1}_3^\top \upsilon_\tau \mathbb{1}_3 - \Delta \sum_{i=1}^{3} \mu_i^{\mathbb{Q}} (\tau - B_\tau^{(i)}),$$
$$B_\tau^{(i)} = \frac{1 - (1 - \kappa_i^{\mathbb{Q}})^\tau}{\kappa_i^{\mathbb{Q}}}, \quad i = 1, 2, 3,$$

where $\mathbb{1}_3$ is the three-dimensional column vector containing ones as elements and υ_τ is a 3×3 matrix whose element on row i and column l, $\upsilon_\tau^{i,l}$, is

$$\upsilon_\tau^{i,l} = \frac{\sigma_i \sigma_l}{\kappa_i^{\mathbb{Q}} \kappa_l^{\mathbb{Q}}} \Gamma_{i,l} \left[\tau - B_\tau^{(i)} - B_\tau^{(l)} + \frac{1 - (1 - \kappa_i^{\mathbb{Q}})^\tau (1 - \kappa_l^{\mathbb{Q}})^\tau}{1 - (1 - \kappa_i^{\mathbb{Q}})(1 - \kappa_l^{\mathbb{Q}})}\right].$$

The model is estimated with maximum likelihood using a Kalman filter on the Canadian end-of-month yield curve data from January 1986 to January 2022. Resulting parameter estimates are given in Table 6.

Table 6: DG3 model parameter estimates

i	κ_i	μ_i	σ_i	λ_i
1	0.00523	0.01780	0.00538	0.72614
2	0.04409	-0.00323	0.00489	-4.76851
3	0.02063	0.05016	0.00810	0.96129

$$\Gamma = \begin{bmatrix} 1 & 0.146 & -0.785 \\ 0.146 & 1 & -0.569 \\ -0.785 & -0.569 & 1 \end{bmatrix}$$

Notes: Parameter estimates for the discrete-time G3 model of Augustyniak et al. (2021) given by Equations (B.1)-(B.2). The maximum likelihood estimation procedure is conducted with a Kalman filter on the Canadian end-of-month yield curve data extending from January 1986 to January 2022.

B.2 Dynamic Nelson-Siegel model

To represent the DNS model dynamics, we consider the same model specification than for the DTAFNS model, except that we that we use $\log A_\tau = 0$ and $B_\tau = \left[\tau,\ \frac{1-e^{-\lambda\tau}}{\lambda},\ \frac{1-e^{-\lambda\tau}}{\lambda} - \tau e^{-\lambda\tau}\right]^\top$ in the bond pricing formula (2.15).

A global optimization of the log-likelihood is conducted with the `R` package `Rsolnp` on the Canadian end-of-month yield curve data extending from January 1986 to January 2022. Corresponding parameter estimates are provided in Table 7.

Table 7: Dynamic Nelson-Siegel model parameters estimates.

i	$\kappa^{\mathbb{P}}_{i,i}$	γ_i	$\Sigma_{i,i}$	$\theta^{\mathbb{P}}_i$	$\theta^{\mathbb{Q}}_i$	$\bar{x}_{i,1\|0}$	λ	h
1	0.0090	1.5184	0.0060	0	0.2000	0.1374		
2	0.0142	1.1746	0.0062	0.0175	0.1912	-0.0351	0.0069	4.4281×10^{-6}
3	0.0226	1.0405	0.0151	0.0686	0.2238	-0.0531		

$$\rho = \begin{bmatrix} 1 & -0.8397 & -0.9202 \\ 0.8397 & 1 & 0.7106 \\ -0.9202 & 0.7106 & 1 \end{bmatrix},\ P_{1|0} = \begin{bmatrix} 4\times 10^{-6} & 0 & 0 \\ 0 & 4\times 10^{-6} & 0 \\ 0 & 0 & 4\times 10^{-6} \end{bmatrix}$$

Notes: Parameter estimates for the dynamic Nelson-Siegel model. The estimation is conducted with the `R` package `Rsolnp`. The data sample consists of Canadian end-of-month yield curve from January 1986 to January 2022.

References

Ang, A., Boivin, J., Dong, S., and Loo-Kung, R. (2011). Monetary policy shifts and the term structure. *The Review of Economic Studies*, 78(2):429–457.

Ang, A. and Piazzesi, M. (2003). A no-arbitrage vector autoregression of term structure dynamics with macroeconomic and latent variables. *Journal of Monetary Economics*, 50(4):745–787.

Augustyniak, M., Godin, F., and Hamel, E. (2021). A mixed bond and equity fund model for the valuation of variable annuities. *ASTIN Bulletin: The Journal of the IAA*, 51(1):131–159.

Bolder, D., Metzler, A., and Johnson, G. (2004). An empirical analysis of the Canadian term structure of zero-coupon interest rates. Staff working paper 2004-48, Bank of Canada.

Brennan, M. J. and Schwartz, E. S. (1979). A continuous time approach to the pricing of bonds. *Journal of Banking & Finance*, 3(2):133–155.

Brigo, D. and Mercurio, F. (2007). *Interest rate models-theory and practice: with smile, inflation and credit*. Springer Science & Business Media.

Chen, R.-R. and Scott, L. (1992). Pricing interest rate options in a two-factor Cox–Ingersoll–Ross model of the term structure. *The Review of Financial Studies*, 5(4):613–636.

Chen, R.-R. and Scott, L. (1993). Maximum likelihood estimation for a multifactor equilibrium model of the term structure of interest rates. *The Journal of Fixed Income*, 3(3):14–31.

Christensen, J. H., Diebold, F. X., and Rudebusch, G. D. (2011). The affine arbitrage-free class of Nelson–Siegel term structure models. *Journal of Econometrics*, 164(1):4–20.

Cox, J. C., Ingersoll, J. E., and Ross, S. A. (1978). A theory of the term structure of interest rates. *Research Paper, Graduate School of Business, Stanford Univ.*

Dai, Q. and Singleton, K. J. (2000). Specification analysis of affine term structure models. *The Journal of Finance*, 55(5):1943–1978.

Diebold, F. X. and Li, C. (2006). Forecasting the term structure of government bond yields. *Journal of Econometrics*, 130(2):337–364.

Duan, J.-C. and Simonato, J.-G. (1999). Estimating and testing exponential-affine term structure models by Kalman filter. *Review of Quantitative Finance and Accounting*, 13(2):111–135.

Duffee, G. R. (2002). Term premia and interest rate forecasts in affine models. *The Journal of Finance*, 57(1):405–443.

Duffie, D. and Kan, R. (1996). A yield-factor model of interest rates. *Mathematical Finance*, 6(4):379–406.

Engle, R., Roussellet, G., and Siriwardane, E. (2017). Scenario generation for long run interest rate risk assessment. *Journal of Econometrics*, 201(2):333–347.

Filipović, D. (1999). A note on the Nelson–Siegel family. *Mathematical Finance*, 9(4):349–359.

Ghalanos, A. and Theussl, S. (2015). *Rsolnp: General Non-linear Optimization Using Augmented Lagrange Multiplier Method*. R package version 1.16.

Gouriéroux, C. and Jasiak, J. (2006). Autoregressive gamma processes. *Journal of Forecasting*, 25(2):129–152.

Guidolin, M. and Timmermann, A. (2006). Term structure of risk under alternative econometric specifications. *Journal of Econometrics*, 131(1-2):285–308.

Heath, D., Jarrow, R., and Morton, A. (1992). Bond pricing and the term structure of interest rates: A new methodology for contingent claims valuation. *Econometrica: Journal of the Econometric Society*, pages 77–105.

Hong, Z., Niu, L., and Zeng, G. (2016). Discrete-time arbitrage-free nelson-siegel term structure model and application. *Available at SSRN 2731041*.

Hördahl, P., Tristani, O., and Vestin, D. (2006). A joint econometric model of macroeconomic and term-structure dynamics. *Journal of Econometrics*, 131(1-2):405–444.

Hull, J. and White, A. (1990). Pricing interest-rate-derivative securities. *The Review of Financial Studies*, 3(4):573–592.

Jamshidian, F. (1996). Bond, futures and option evaluation in the quadratic interest rate model.

Applied Mathematical Finance, 3(2):93–115.

Kalman, R. E. (1960). A new approach to linear filtering and prediction problems. *Transactions of the ASME–Journal of Basic Engineering*, (Series D)(82):35–45.

Le, A., Singleton, K. J., and Dai, Q. (2010). Discrete-time affine$^{\mathbb{Q}}$ term structure models with generalized market prices of risk. *The Review of Financial Studies*, 23(5):2184–2227.

Lemke, W. (2006). *Term structure modeling and estimation in a state space framework*, volume 565. Springer Science & Business Media.

Litterman, R. and Scheinkman, J. (1991). Common factors affecting bond returns. *Journal of Fixed Income*, 1(1):54–61.

Nelson, C. R. and Siegel, A. F. (1987). Parsimonious modeling of yield curves. *Journal of Business*, pages 473–489.

Nelson, D. B. (1991). Conditional heteroskedasticity in asset returns: A new approach. *Econometrica: Journal of the Econometric Society*, pages 347–370.

Park, H. (2014). Estimation of affine term structure models under the Milstein approximation. *Applied Economics Letters*, 21(9):651–656.

Rogers, L. (1995). Which model for term-structure of interest rates should one use? *Institute for Mathematics and Its Applications*, 65:93.

Rémillard, B. (2013). *Statistical methods for financial engineering*. CRC press.

Shumway, R. H. and Stoffer, D. S. (2017). *Time series analysis and its applications. With R examples*, volume 4. Springer Texts in Statistics.

Svensson, L. E. (1994). Estimating and interpreting forward interest rates: Sweden 1992-1994.

Vasicek, O. (1977). An equilibrium characterization of the term structure. *Journal of Financial Economics*, 5(2):177–188.

Wüthrich, M. V. and Merz, M. (2013). *Financial modeling, actuarial valuation and solvency in insurance*. Springer.

Ye, Y. (1987). *Interior Algorithms for Linear, Quadratic, and Linearly Constrained Non-Linear Programming.* PhD thesis, Department of ESS, Stanford University.

Online Appendix (not part of the paper)

C Kalman filter

Filtering techniques are statistical methods applied to perform inference on latent variables, referred to as the signal, based on observations linked to such signal. The most popular filtering method is the Kalman filter which can be applied under the assumption of linear and Gaussian dynamics for the evolution of the signal and the relationship between the signal and observations. More precisely, as presented in Rémillard (2013), the such dynamics for the signal $\{Z_i\}_{i=1}^n$ and observations $\{Y_i\}_{i=1}^n$ is represented by

$$
\begin{aligned}
Z_i &= \mu_i + F_i Z_{i-1} + G_i w_i, \\
Y_i &= d_i + U_i Z_i + \epsilon_i,
\end{aligned}
$$

where $\mu_i \in R^m$, $F_i \in R^{m\times m}$, $G_i \in R^{m\times q}$, $d_i \in R^r, U_i \in R^{r\times m}$, $w_i \sim N_q(0, W_i)$ and $\epsilon_i \sim N_r(0, R_i)$. Independent processes $\{w_i\}_{i=1}^n$ and $\{\epsilon_i\}_{i=1}^n$ are sequences of independent and identically distributed random variables. For $i = 1, \ldots, n$, set

$$
\begin{aligned}
Y_{i|i-1} &= \mathbb{E}[Y_i|Y_{i-1}, ..., Y_1], \\
Z_{i|i-1} &= \mathbb{E}[Z_i|Y_{i-1}, ..., Y_1], \\
Z_{i|i} &= \mathbb{E}[Z_i|Y_i, ..., Y_1], \\
P_{i|i-1} &= \mathbb{E}[(Z_i - Z_{i|i-1})(Z_i - Z_{i|i-1})^\top|Y_{i-1}, ..., Y_1], \\
P_{i|i} &= \mathbb{E}[(Z_i - Z_{i|i})(Z_i - Z_{i|i})^\top|Y_i, ..., Y_1], \\
e_i &= Y_i - Y_{i|i-1}, \\
V_i &= \mathbb{E}[e_i e_i^T|Y_{i-1}, ..., Y_1].
\end{aligned}
\tag{C.1}
$$

Then, the following proposition holds, see Rémillard (2013).

Proposition C.1. *Suppose that $Z_0 \sim N_m(Z_{0|0}, P_{0|0})$. Given $Y_1, \ldots, Y_i$, the conditional distribu-*

tion

- *of Z_i is Gaussian, with mean $Z_{i|i}$ and covariance matrix $P_{i|i}$,*
- *of Z_{i+1} is Gaussian with mean $Z_{i+1|i}$ and covariance matrix $P_{i+1|i}$,*
- *of Y_{i+1} is Gaussian with mean $Y_{i+1|i}$ and covariance matrix V_{i+1}.*

Kalman (1960) provides a recursive algorithm to calculate conditional moments of the signal and observations outlined in Proposition C.1. For given initial moments $Z_{0|0}$ and $P_{0|0}$, the algorithm goes as follows for $i = 1, \ldots, n$:

$$\begin{aligned}
Z_{i|i-1} &= \mu_i + F_i Z_{i-1|i-1}, \\
Y_{i|i-1} &= d_i + U_i Z_{i|i-1}, \\
P_{i|i-1} &= F_i P_{i-1|i-1} F_i^\top + G_i W_i G_i^\top, \\
V_i &= U_i P_{i|i-1} U_i^\top + R_i, \\
Z_{i|i} &= Z_{i|i-1} + P_{i|i-1} U_i^\top V_i^{-1} e_i, \\
P_{i|i} &= P_{i|i-1} - P_{i|i-1} U_i^\top V_i^{-1} U_i P_{i|i-1}.
\end{aligned}$$

Using previous results and calculations, the joint distribution of $Y_1, ..., Y_n$ can be computed, and the corresponding log-likelihood L is given by

$$-L = -\sum_{i=1}^{n} \log p(Y_i|Y_1, \ldots, Y_{i-1}) = \frac{1}{2}\sum_{i=1}^{n} \log\left(\det V_i\right) + \frac{1}{2}\sum_{i=1}^{n} e_i^\top V_i^{-1} e_i + \frac{nr}{2}\ln(2\pi).$$

To estimate the parameters of the DTAFNS model, the following substitutions need to be applied in the above Kalman filter algorithm: $Y_i = \hat{\mathsf{y}}_i$, $Z_i = X_i$, $i = t$, $r = M$, $m = q = 3$, $\mu_i = \mathbf{b} = \kappa^{\mathbb{P}}\theta^{\mathbb{P}}$, $F = \mathbf{D} = (I - \kappa^{\mathbb{P}})$, $G_i = \Sigma$, $d_i = \mathsf{a}$, $U_i = \mathsf{B}$, $R_i = H$ and $W_i = \rho$.

D Alternative parameters selection for DTAFNS

As explained in Remark 2.2, it might be possible to consider a specification of the DTAFNS model under which $\kappa^{\mathbb{P}}_{1,1} = 0$, which (i) would be consistent with $\kappa^{\mathbb{Q}}_{1,1} = 0$ and (ii) would also yield non-stationary dynamics for the first factor under physical dynamics. From (2.17), setting $\kappa^{\mathbb{P}}_{1,1} = 0$ leads to $\gamma_1 = 0$. Such specification is tested in this appendix. Table 8 shows the parameter estimation results under such additional constraints.

Table 8: Maximum likelihood estimates of the DTAFNS model with constraint $\kappa^{\mathbb{P}}_{1,1} = 0$

$$\rho = \begin{bmatrix} 1 & -0.6300 & -0.4101 \\ -0.6300 & 1 & 0.2991 \\ -0.4101 & 0.2991 & 1 \end{bmatrix}$$

i	$\kappa^{\mathbb{P}}_{i,i}$	γ_i	$\Sigma_{i,i}$	$\theta^{\mathbb{P}}_i$	$\theta^{\mathbb{Q}}_i$	λ	h
1	0	0	0.0027	0	0		
2	0.2966	1.3807	0.0046	0.0501	0.0637	0.0233	3.76×10^{-6}
3	0.0361	1.8247	0.0070	0.0495	0.0767		

Notes: Maximum likelihood parameters estimates for the DTAFNS model presented in Section 2.2 obtained with the Kalman filter-based Algorithm 1. The data sample is the Canadian end-of-month yield curve data extending from January 1986 to January 2022.

Table 9 shows the in-sample and out-of-sample log-likelihoods of both the original version of the DTAFNS model from Section 2.2 and the version with the added constraint $\kappa^{\mathbb{P}}_{1,1} = 0$. Imposing $\kappa^{\mathbb{P}}_{1,1} = 0$ deteriorates the results both in-sample and out-of-sample, which justified ultimately not retaining this specification.

Table 9: Log-likelihood of the DTAFNS model with and without constraint $\kappa^{\mathbb{P}}_{1,1} = 0$

Model	Out-of-sample						In-sample
	2017	2018	2019	2020	2021-22	Aggregated	
DTAFNS ($\kappa^{\mathbb{P}}_{1,1} \neq 0$)	2,034	2,001	2,035	2,015	2,183	10,268	65,702
DTAFNS ($\kappa^{\mathbb{P}}_{1,1} = 0$)	2,005	1,977	2,015	2,002	2,175	10,174	65,700

Notes: Comparison of the log-likelihood of the DTAFNS model of Section 2.2 and a version of the same model with the added constraint $\kappa^{\mathbb{P}}_{1,1} = 0$. The data sample consists of Canadian end-of-month yield curves. The in-sample dataset starts in January 1986 and ends in January 2022. The out-of-sample estimation procedure uses an expanding window approach described in Section 3.3.2. The aggregated out-of-sample log-likelihood is obtained by summing the log-likelihoods for all test years.

Table 10: Probability of observing negative short rates with the DTAFNS model with and without constraint $\kappa^{\mathbb{P}}_{1,1} = 0$

Model	Proportion of months				Proportion of paths with at least one month			
	< 0	< -0.01	< -0.02	< -0.03	< 0	< -0.01	< -0.02	< -0.03
DTAFNS($\kappa^{\mathbb{P}}_{1,1} \neq 0$)	0.177	0.057	0.015	0.003	0.644	0.299	0.106	0.029
DTAFNS($\kappa^{\mathbb{P}}_{1,1} = 0$)	0.322	0.141	0.052	0.016	0.826	0.540	0.277	0.113

Notes: Within 200,000 five-year monthly simulated paths of the DTAFNS model (either with or without constraint $\kappa^{\mathbb{P}}_{1,1} = 0$), the proportion of simulated months with short rates being smaller than respectively 0, -0.01, -0.02 and -0.03, and the proportion of simulated paths with at least one month below such thresholds. Model parameters estimates are from Table 8 and Table 1, with starting values of the factors being smoothed factor values on January 2022 associated with such sets of parameters.

E Point prediction performance for spot rates

In this section, we compare the ability of the various considered models to produce accurate point monthly forecasts for spot rates. Let $\hat{y}(t, t+\tau)$ be the realized time-t spot rates with tenor τ, and $\tilde{y}(t, t+\tau)$ be its associated model-implied forecast produced at time $t-1$ with the expanding window approach detailed in Section 3.3.2, i.e. models are retrained yearly, with yearly out-of-sample test sets covering the period extending from January 2017 to January 2022. Define the following performance metrics:

$$\begin{aligned}
\text{Mean error} =& \frac{1}{T^*} \sum_{t=1}^{T^*} (\hat{y}(t, t+\tau) - \tilde{y}(t, t+\tau)), \\
RMSE^{(\tau)} =& \sqrt{\frac{1}{T^*} \sum_{t=1}^{T^*} (\hat{y}(t, t+\tau) - \tilde{y}(t, t+\tau))^2}, \\
MAE^{(\tau)} =& \frac{1}{T^*} \sum_{t=1}^{T^*} |\hat{y}(t, t+\tau) - \tilde{y}(t, t+\tau)|, \\
rMAE_1^{(\tau)} =& \frac{MAE^{(\tau)}_{DTAFNS}}{MAE^{(\tau)}_{DG3}}, \quad rMAE_2^{(\tau)} = \frac{MAE^{(\tau)}_{DTAFNS}}{MAE^{(\tau)}_{DNS}}
\end{aligned} \tag{E.1}$$

where T^* is the total number of observations in the union of all test sets and $MAE^{(\tau)}_{\mathcal{M}}$ is the $MAE^{(\tau)}$ for model $\mathcal{M}$. Table 11 reports the predictive performance results. RMSE and MAE metrics indicate that the point forecast performance of the DTAFNS model is quite similar to that of the DG3 and DNS models.

Table 11: Performance metrics for spot rate point predictions

Tenor (months)	Mean error (in %)			RMSE (in %)			MAE (in %)			rMAE	
	DTAFNS	DG3	DNS	DTAFNS	DG3	DNS	DTAFNS	DG3	DNS	$rMAE_1$	$rMAE_2$
6	0.048	0.075	0.120	0.707	0.712	0.734	0.612	0.609	0.620	1.004	0.986
12	0.155	0.160	0.166	0.731	0.732	0.742	0.629	0.630	0.634	0.999	0.992
36	0.089	0.054	-0.009	0.661	0.651	0.648	0.527	0.517	0.513	1.019	1.027
60	-0.043	-0.072	-0.137	0.626	0.623	0.628	0.472	0.472	0.484	1.000	0.975
84	-0.186	-0.192	-0.237	0.632	0.631	0.640	0.476	0.477	0.489	1.000	0.975
108	-0.296	-0.280	-0.298	0.652	0.644	0.647	0.484	0.477	0.480	1.014	1.009
132	-0.375	-0.346	-0.337	0.671	0.657	0.648	0.491	0.479	0.469	1.026	1.048
156	-0.433	-0.400	-0.366	0.689	0.671	0.648	0.500	0.483	0.463	1.035	1.081
180	-0.472	-0.443	-0.390	0.701	0.683	0.649	0.510	0.493	0.462	1.034	1.103
204	-0.494	-0.473	-0.408	0.704	0.692	0.649	0.517	0.505	0.465	1.024	1.113
228	-0.498	-0.489	-0.422	0.699	0.694	0.649	0.517	0.511	0.469	1.011	1.102
252	-0.488	-0.492	-0.434	0.684	0.688	0.648	0.505	0.508	0.472	0.993	1.069
276	-0.466	-0.481	-0.445	0.662	0.673	0.648	0.485	0.496	0.475	0.979	1.021
300	-0.433	-0.457	-0.456	0.633	0.650	0.649	0.465	0.478	0.477	0.972	0.974
324	-0.389	-0.417	-0.466	0.599	0.618	0.651	0.441	0.455	0.482	0.969	0.913
348	-0.331	-0.361	-0.476	0.559	0.577	0.652	0.415	0.426	0.489	0.974	0.848
Average	-0.286	-0.286	-0.287	0.661	0.660	0.658	0.502	0.501	0.497	1.003	1.011

Notes: Performance metrics for the point forecasting of spot rates by the DTAFNS model and its two benchmarks explained in Appendix B. The expanding window approach detailed in Section 3.3.2 is used for testing: models are retrained yearly, with yearly out-of-sample test sets covering the period extending from January 2017 to January 2022. The rMAE metric is explained in (E.1). Performance metrics are reported for a selection of 16 tenors (in months) among the 33 available in dataset, while the average row at the bottom considers all tenors available.

F Fit diagnostics output for the mixed fund model

In this section, diagnostic information about mixed fund parameter estimates is presented. Parameters mu, mxreg1, mxreg2, mxreg3, mxreg4, mxreg5, mxreg6 in the output below correspond to ψ_0, ψ_1, ψ_2, ψ_3, ψ_3', $\psi_1^{(S)}$, and $\psi_2^{(S)}$, respectively. Results of several white noise tests are also presented to verify the adequacy of the model; weighted Ljung-Box tests on standardized residuals and squared residuals, and weighted ARCH Lagrange Multiplier (LM) tests are conducted to assess the presence of auto-correlation and volatility clusters in residuals. See the `rugarch` package documentation for details of specifications of such tests.

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.004039	0.001227	3.2907	0.000999
mxreg1	−1.724464	0.404332	−4.2650	0.000020
mxreg2	−0.769503	0.242516	−3.1730	0.001509
mxreg3	−0.445368	0.111513	−3.9939	0.000065
mxreg4	−0.055235	0.019353	−2.8540	0.004317
mxreg5	0.400758	0.014636	27.3815	0.000000
mxreg6	0.134577	0.013903	9.6800	0.000000
omega	−1.031221	0.012377	−83.3193	0.000000
alpha1	−0.093903	0.035272	−2.6623	0.007761
beta1	0.894535	0.000279	3205.6931	0.000000
gamma1	0.066528	0.020912	3.1814	0.001466

LogLikelihood : 1038.719

Weighted Ljung–Box Test on Standardized Residuals

---

	statistic	p–value
Lag[1]	0.6482	0.4208
Lag[2∗(p+q)+(p+q)−1][2]	1.3704	0.3922
Lag[4∗(p+q)+(p+q)−1][5]	2.3021	0.5490

d.o.f=0

H0 : No serial correlation

Weighted Ljung–Box Test on Standardized Squared Residuals

---

	statistic	p–value
Lag[1]	0.3453	0.5568
Lag[2*(p+q)+(p+q)−1][5]	1.7040	0.6898
Lag[4*(p+q)+(p+q)−1][9]	5.8407	0.3169

d.o.f=2

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P–Value
ARCH Lag[3]	0.8588	0.500	2.000	0.3541
ARCH Lag[5]	2.7687	1.440	1.667	0.3251
ARCH Lag[7]	4.7279	2.315	1.543	0.2535